

Online Appendix for “Natural Capital and the Measurement of Productivity”*

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July 12, 2026

*All errors are our own.

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1 The Theory of Natural Capital and Total Factor Productivity in Growth Accounting

This appendix examines the conditions under which incorporating a previously unmeasured natural capital input into a macroeconomic production function improves residual-based estimates of total factor productivity (TFP). Our laboratory for measurement is a theoretical economy in which output is governed by TFP as well as by the inputs to the aggregate production function F . We assume that F is log-linear in inputs and takes at least four factors as arguments: labor L , produced capital K , and $J > 1$ natural capital inputs indexed $\{N_{j=1}, N_{j=2}, \dots, N_{j=J}\}$.¹ The logarithm of output at each instant t is given by

$$\begin{aligned} \ln(Y(t)) &= \ln\left(TFP(t) \times F(K(t), L(t), \{N_j(t)\}_{j=1}^J)\right) \\ &= \ln(TFP(t)) + \gamma_K(t) \ln(K(t)) + \gamma_L(t) \ln(L(t)) + \sum_{j=1}^J \gamma_j(t) \ln(N_j(t)) \end{aligned} \quad (1)$$

where $\gamma_K, \gamma_L, \{\gamma_j\}_{j=1}^J > 0$ denote output elasticities with respect to the corresponding factor inputs. If F is homogeneous of degree one in its inputs (i.e., production exhibits constant returns to scale, CRS), output can be expressed as

$$Y(t) = TFP(t) \times \left(F_K K(t) + F_L L(t) + \sum_{j=1}^J F_{N_j} N_j(t)\right) \quad (2)$$

where F_x denotes the partial derivative of F with respect to argument $x \in \{K, L, \{N_j\}_{j=1}^J\}$. Under the assumption that the output elasticities γ are time-invariant, the growth rate of output in (1) at a given time t can be expressed as²

$$\frac{\dot{Y}(t)}{Y(t)} = \frac{T\dot{F}P(t)}{TFP(t)} + \frac{\dot{F}(t)}{F(t)} = \frac{T\dot{F}P(t)}{TFP(t)} + \gamma_K \frac{\dot{K}(t)}{K(t)} + \gamma_L \frac{\dot{L}(t)}{L(t)} + \sum_{j=1}^J \gamma_j \frac{\dot{N}_j(t)}{N_j(t)} \quad (3)$$

where dots denote derivatives of a given variable with respect to time. Let $g_X(t) \triangleq \dot{X}(t)/X(t)$ denote the growth rate of variable X at time t for $X \in \{TFP, K, L, \{N_j\}_{j=1}^J\}$. Rearranging (3) yields the accounting-based formula for calculating TFP growth: g_{TFP} is output growth less an output-elasticity weighted sum of growth in inputs:

$$g_{TFP} = g_Y - \gamma_K g_K - \gamma_L g_L - \sum_{j=1}^J \gamma_j g_{N_j} \quad (4)$$

We refer to (4) as the *growth accounting* equation going forward. The growth accounting equation states that if output elasticities in the production function are known to the econometrician and

1. Without loss of generality, other omitted inputs that are not natural capital can be subsumed into the set of J inputs beyond produced capital and labor.

2. Equation (3) still holds when output elasticities depend on the level of inputs and/or time, but they must be time-indexed.

she observes growth in inputs used and output, then she can recover the full path of TFP growth.

Several barriers complicate the inclusion of natural capital (NK) stocks in empirical growth accounting exercises. First, a national accountant may not know the full set of natural capital stocks $\{N_j\}_{j=1}^J$ that enter the production function. National statistical agencies have not traditionally collected data on natural capital. The System of National Accounts (SNA) standard international guidance includes a set of natural assets, but few countries tabulate changes in these assets over time (United Nations 2025). The System of the Environmental-Economic Accounts Central Framework and Ecosystem Accounts provides guidance and additional natural assets countries are expected to track (United Nations et al. 2014). While some countries have begun compiling natural capital accounts, these data seldom capture the full scope of natural assets that may enter the production function. Second, the accountant cannot (at least directly) observe output elasticities γ . Elasticities must be estimated, which introduces uncertainty and potential bias when these elasticities are used in the growth accounting equation. Third, the accountant may face measurement error when calculating growth in output and factor inputs as well as any other auxiliary measurements required to estimate output elasticities.

This section presents a theoretical framework for examining how uncertainty over which natural capital stocks enter the production function affects estimates of output elasticities. The difficulties facing estimation and measurement we show here enumerate the barriers to measurement highlighted in the text of the main manuscript. We show conditions under which more information about some, but not all, inputs sharpens estimates of output elasticities. These are policy-relevant questions; it is unlikely that national accountants will go from current practices to identifying the complete set of NK assets without first identifying an incomplete set.

1.1 Estimating Output Elasticities

Estimating TFP growth using the growth accounting method requires estimates of the output elasticities in (4). There are two general approaches; each requires a distinct set of assumptions. The first approach, the income approach, relaxes the assumption of log-linearity and operates under two alternative assumptions: F is assumed to exhibit constant returns to scale, (2), and the structures of all markets are known. The second approach, the regression approach, requires only that the aggregate production function in the economy is log-linear as in (1). This section briefly reviews the income approach before giving an overview of the regression approach as well as its properties when natural capital is omitted from the growth accounting equation.

1.2 Income-Based Approach

When the production function F exhibits CRS and factor markets are perfectly competitive, then in equilibrium, the shares of national income accruing to each factor input are equal to their output elasticities (Caselli and Feyrer 2007). For the case of capital, this implies

$$\gamma_K = \frac{F_K K}{Y} = \frac{RK}{Y} \quad (5)$$

The ratio of total rents paid to capital, RK , divided by output Y , often measured as gross domestic product (GDP), is an unbiased estimator of the capital output elasticity in the economy. This procedure is useful because its arguments can be measured in *nominal* terms as opposed to real terms. Analogous procedures can be used to estimate elasticities for labor and natural capital factors where the factor income accruing to them is observable. An added advantage is that under the CRS assumption, factor shares of national income must sum to one.

These estimators become biased if any factor income attributable to an omitted factor, e.g., any natural capital stock that enters production but is not paid its factor income, is misattributed to observed factors. Manning, Taylor, and Wilen (2018) illustrate how this case is likely for industries that use rival public goods as inputs (the authors there use a fishery as a running example) and show how it complicates measurement of factor shares in production. The estimator in (5) is biased if the aggregate production function does not exhibit CRS or the econometrician does not know the structure governing the market for factor inputs. The challenges of estimating factor shares using this method are well known (Caselli and Feyrer 2007; Hsieh and Klenow 2010; Monge-Naranjo, Sánchez, and Santaaulalia-Llopis 2019). The resulting measurement error when using the income-based approach to calculate TFP growth depends on how national income is measured, how national output is attributed across factors of production, and how the error propagates through to TFP calculations.

1.3 Regression-Based Estimators

The regression-based approach relies on the log-linear restriction in (1). When production is Cobb-Douglas or well-approximated by its Taylor expansion such as in the translog case, the first two moments of the stochastic process governing input and output growth are sufficient statistics for the set of output elasticities. The regression approach remains valid even when markets are not perfectly competitive as long as *quantities* (as opposed to value products) of inputs and output are known.³

To fix ideas, consider the running example from the main text, an economy that uses two natural capital inputs, N_1 and N_2 , with output elasticities γ_1 and γ_2 , in addition to produced capital and labor. Suppose a national accountant observes time-series data on changes in output, g_Y (i.e., changes in real GDP), along with growth in produced capital g_K and labor g_L , and measures all quantities she observes without error. Assume from here forward that all measured factors are measured without error. An econometrician who observes g_K and labor g_L has sufficient data to estimate γ_K and γ_L using a regression model based on the presupposed data-generating process

$$g_Y = \hat{g}_A + \hat{\gamma}_K g_K + \hat{\gamma}_L g_L + \epsilon \quad (\text{Model 1})$$

3. This condition is rarely met in practice because output is typically measured in terms of monetary value as opposed to physical quantities.

where $\hat{g}_A$ is an intercept term that becomes an estimator of average TFP growth. However, equation (3) implies that the true data-generating process (DGP) is

$$g_Y = \gamma_K g_K + \gamma_L g_L + \gamma_1 g_{N_1} + \gamma_2 g_{N_2} + g_{TFP} \quad (\text{DGP})$$

If one estimates (Model 1) by ordinary least squares (OLS), the fitted coefficients will be subject to traditional omitted variable bias (OVB) (Hansen 2022). For example, when $\hat{\gamma}_K$ is the OLS estimand from Model 1, it identifies

$$\hat{\gamma}_K - \gamma_K = \gamma_1 \kappa_{N_1, \tilde{K}} + \gamma_2 \kappa_{N_2, \tilde{K}} + \kappa_{TFP, \tilde{K}} \quad (\text{Bias 1})$$

where $\kappa_{X, \tilde{K}}$ is the coefficient from a linear projection of the growth of X onto the residualized growth in K . By (Bias 1), unless growth in capital, K , and in labor, L , are uncorrelated with growth in the omitted natural capital factor inputs and TFP, estimates of $\hat{\gamma}_K$ and $\hat{\gamma}_L$ from Model 1 will be biased. The magnitude of the bias depends on the magnitude of the output elasticities for omitted factors and on how they covary with observed factors.

Consider an alternative case where one of the productive natural capital stocks, N_1 , is observed by the econometrician. The analyst may expect that including this additional factor will reduce the bias in the estimates. They consider estimating

$$g_Y = \tilde{g}_A + \tilde{\gamma}_K g_K + \tilde{\gamma}_1 g_{N_1} + \tilde{\gamma}_L g_L + \nu \quad (\text{Model 2})$$

where tildes differentiate Model 2 from Model 1. The bias in the estimate of the capital output elasticity using (Model 2) is

$$\tilde{\gamma}_K - \gamma_K = \gamma_2 \lambda_{N_2, \tilde{K}} + \lambda_{TFP, \tilde{K}} \quad (\text{Bias 2})$$

where the λ s are coefficients from residualized auxiliary regressions as above. It is ambiguous whether (Bias 1) or (Bias 2) is greater without further assumptions.

Consider omitted variable bias when g_{N_1} and g_K are the only terms in the growth accounting equation that are not iid. The predictions of theory are immediate; OVB is present in model 1 only, and including g_{N_1} always (weakly) reduces bias when estimating γ_K . Now consider relaxing this assumption slightly, such that $\rho(g_{N_1}, g_{TFP}) > 0$ while g_{TFP} remains uncorrelated with all other growth terms (including g_K).⁴ In this case, models 1 and 2 produce biased estimates of γ_K because $\kappa_{N_1, \tilde{K}}$ and $\lambda_{N_1, \tilde{K}}$ are nonzero, making it ambiguous which estimator is less biased. For Model 1, the bias is determined by $\kappa_{N_1, \tilde{K}}$, which is the correlation between g_{N_1} and the portion of the growth in capital that is not explained by growth in labor:

$$\hat{\gamma}_K - \gamma_K = \gamma_1 \kappa_{N_1, \tilde{K}} \quad (\text{Bias 1'})$$

When the correlation between growth in N_1 and the growth in produced capital not explained by

4. The choice of direction for the inequality here is arbitrary and symmetric results would follow if it were reversed.

growth in other factors is positive ($\kappa_{N_1, \tilde{K}} > 0$), some of the growth in output caused by growth in N_1 is misattributed to produced capital, leading to an overestimate of γ_K . In Model 2, the bias is given by:

$$\tilde{\gamma}_K - \gamma_K = \lambda_{A, \tilde{K}} \quad (\text{Bias 2'})$$

and will lead to an RMSE that depends on the magnitude of the correlation between growth in N_1 and growth in TFP. The bias in (Bias 2') has the opposite sign of the correlation between growth in N_1 and TFP because the regression coefficient $\lambda_{A, \tilde{K}}$ is the covariance in TFP and the portion of capital growth *not* explained by g_{N_1} . Since g_K and g_{TFP} are *unconditionally* uncorrelated, as g_{TFP} is positively correlated with g_{N_1} and a portion of g_K that is correlated with g_{N_1} , the bias results from the negative correlation with the portion of g_K that is orthogonal to g_{N_1} . Overall, this leads to a correlation that is zero. The result is that g_{TFP} is negatively correlated with $\tilde{K}$, the residualized portion of g_K , so that γ_K is underestimated when $\rho(g_{N_1}, g_{TFP}) > 0$.

2 Empirical Methodology and Additional Results

This section gives additional information on the empirical methodology as well as some robustness checks for the regression model in Section 2 of the manuscript.

2.1 Data for Covered Countries in the Empirical Portion of the Main Text

We estimate equation (13) from the main text on a balanced panel of 117 countries over the 1996-2019 period. Table 1 displays the list of countries covered by estimating sample. Due to the paucity of data spanning the entire panel for some resources, we impute zeros to the growth rates for all resources where the observation in the prior year is a missing value. This imposes the assumption that there is no change in the true usage rate of natural services where data from the prior year are missing. The last column shows the share of data points that are imputed values of zero for each resource. Outside of mangroves, biomass, and hydropower, the imputation is minimal; however, for the latter three resources a large portion of the original data are missing entries.

2.2 Robustness Checks

We turn here to present a number of robustness checks of our empirical exercise in the main text. Equation (5) in the main text shows that we cannot differentiate between correlation natural capital stock growth and TFP growth driven by omitted environmental inputs and that driven by other forms of omitted variable bias. This appendix attempts to bolster our findings by showing that the results are driven by our choice of outcome variables or the natural capital stocks we include as regressors.

2.2.1 Additional Results for Main Specification

Table 2 in the main text estimates nested variations of

$$g_{A_{j,t}} = \sum_{s \in S} \beta_s g_{S_{s,j,t}} + \theta' \mathbf{Z}_{j,t} + \delta_j + \zeta_t + \zeta_j t + \varepsilon_{j,t}, \quad (6)$$

where, in the specification in column (5), $\mathbf{Z}_{j,t}$ is a set of controls from the PWT, δ_j and ζ_t are two-way fixed effects, and ζ_j are country-specific idiosyncratic trend terms. These specifications, while standard, are chosen on an ad hoc basis. To ensure we are not spuriously rejecting the null hypothesis due to our choice of specifications, we implement the post double selection (PDS) LASSO approach proposed in Belloni, Chernozhukov, and Hansen (2014) (BCH) to take a systematic approach to including controls. The PDS procedure uses a LASSO to select controls that are strong predictors of the outcome variable and covariates of interest (i.e., growth in the six natural capital stocks) in order to improve the precision on estimates of interest while preserving important controls. This provides a systematic method for selecting which fixed effects, controls, and trend terms to include in the specification as opposed to the benchmark nesting approach.

The first row of Table 3, labeled “PDS” in the second column, shows p -values from Wald and wild bootstrap (Roodman et al. 2019) tests of the joint null hypothesis $\beta = 0$ when the specification is chosen using the PDS LASSO. The second and third rows show the results from when the null hypothesis is tested under the initial LASSO specification (“LASSO”) or during the intermediate step using the residuals from the second stage of the BCH procedure (“post-LASSO”). The Wald tests use robust standard errors (again, due to the paucity of effective clusters) while the wild bootstrap tests are clustered at the country level. We can reject the null hypothesis of no correlation between natural capital stock growth and measured TFP growth at the 1 percent level under all specifications for both tests.

Rows 4–6 of Table 3 estimate versions of (6) and use Driscoll and Kraay (1998) standard errors to allow for cross-sectional as well as two years’ serial dependence in error terms. All models include two-way fixed effects, while rows 5 and 6 add controls $\mathbf{Z}_{j,t}$ and idiosyncratic trend terms. We can reject the joint null hypothesis of no correlation under both Wald and wild bootstrap tests at standard levels.

2.2.2 Randomization Inference Tests of Joint Significance

It is possible that individual observations with high leverage are creating spurious results of joint significance due to the imputation process for creating a balanced panel across countries. To check for this, we perform a randomization inference test on the specification in column (3) of Table 2 in the main text. We resample growth in the six natural capital stocks, $g_{S_{s,j,t}}$, within each country without replacement and randomly reassign them to years within the panel. This process, by construction, imposes that the correlation between these newly reassigned regressors $\tilde{g}_{S_{s,j,t}}$ and the outcome variable $g_{A_{j,t}}$ is expected to be zero within each country. We repeat this process 100,000 times. For each permuted dataset, we estimate the following regression

$$g_{A_{j,t}} = \sum_{s \in \mathcal{S}} \beta_s \tilde{g}_{S_{s,j,t}} + \theta' \mathbf{Z}_{j,t} + \delta_j + \zeta_t + \varepsilon_{j,t} \quad (7)$$

and perform a Wald test of joint significance for coefficients β each time. Because the data are constructed such that the null hypothesis for the joint Wald test is true, this test forms an empirical distribution for F -test statistics under the null hypothesis. Figure 1 shows the empirical distribution of test statistics when the null hypothesis is true for 100,000 resampled versions of our panel. The p -value from this test—the share of placebo F -test statistics that are larger than the one in our estimating sample ($F = 2.505$)—is 0.0185.

2.2.3 Choice of Instruments for Natural Capital Services

The six natural capital stocks in Table 2 comprise the baseline set we test because these are the six natural capital stocks in the CWON dataset for which quantity data are public. However, as the choice to include all six is still somewhat arbitrary given they are ultimately only a subset of

renewable natural capital, here we ensure that our rejection of the joint null result is not driven by our choice of instruments.

Excluding Agricultural, and Timber Land Table 4 displays the results when equation 4 in the main text is re-estimated excluding all forms of land resources outside of forests from the set of renewable natural capital stocks. This avoids any bias induced by land resources if they are correlated with measurement error in capital or labor income in national accounts that is used to estimate TFP growth in the PWT. Wald tests of joint nullity using asymptotic standard errors and semiparametric methods become marginal or insignificant at traditional thresholds for statistical significance under most specifications we consider. As discussed in the main text, some of this loss in precision may be due to the large share of missing values for factor growth we impute as zeros here. However, both tests of the joint null are significant at standard levels ($p < 0.05$) when using either the PDS Lasso for covariate selection or when Driscoll and Kraay (1998) standard are errors for either Wald test.

Inclusion of Non-Renewables The analysis in the main text looked at the correlation between TFP growth and changes in stocks of renewable resources as proxies for changes in services from natural capital. It is possible that the significant results there were due to omitted variable bias from leaving out non-renewable resources. Table 5 repeats the exercise in Table 2 of the main text when we include all natural capital flows that can be calculated from the public version of the CWON dataset (i.e., urban land, renewable natural capital, fossil fuels, and subsurface minerals). The full set of non-renewable and renewable natural capital stocks included in this broader specification, in addition to urban land, are listed in Tables 12 and 13. Table 5 shows the estimated coefficients associated with each natural capital stock when equation 13 in the main text is estimated on an unbalanced panel of 118 countries between 1972 and 2019 after expanding $\mathbf{S}$ to the full set of 20 natural capital stocks. Table 6 re-estimates this specification for the 1996-2019 period so it matches the time coverage in the main estimating sample.⁵

Overall, including the full suite of natural capital covariates available in the CWON dataset does not change the qualitative results. Reassuringly, the coefficients on oil (a resource whose associated industries comprise a substantial share of global GDP) are positive and significant at traditional levels in both tables. Agricultural land and hydropower remain precisely estimated. More generally, the coefficients on most natural capital stocks remain positive, in line with our priors that the use of non-renewable resources in production should be “goods” in terms of increasing output. While the coefficient on forest cover is often negative, it is not significant at traditional levels and qualitatively aligned with some specifications shown in the main text. For these specifications including a broader suite of natural capital, we adjust the null hypothesis to include all 20 natural

5. The resulting number of observations and covered countries in this appendix does not coincide with that in the main text as here we adopt a more aggressive imputation strategy in order to fill in missing data. The process for constructing these quantity changes prior to 1996 is described in Section 3. Results are qualitatively unchanged when the estimating sample here is restricted to the 1996-2019 period, as shown in Table 6.

capital stocks. Wald tests using both asymptotic and semiparametric methods become substantially tighter around zero in all specifications we consider.

2.2.4 Results for Alternative Outcome Variables

This section examines whether we find similar correlations between natural capital and productivity growth when we consider alternative measurements of productivity growth to those from the Penn World Tables. We first consider two alternative panels measuring national productivity: one from Eurostat covering the 27 countries in the European Union (plus Norway) that measures economy-wide multi-factor productivity (European Commission and Eurostat 2025), and a second covering only the agricultural sector produced by the Economic Research Service at the USDA (Fuglie 2015). Next, we examine the correlations between natural capital and growth in measured output after residualizing out growth explained by labor and produced capital inputs.

Eurostat MFP Table 7 displays the results when equation 13 in the main text is re-estimated on multifactor productivity series generated by the European Statistical Agency (Eurostat) for the 27 EU countries along with Norway (European Commission and Eurostat 2025). The panel covers the years 2003-2019. Results from this alternative specification are shown in Table 7. While Wald tests of joint nullity using asymptotic standard errors remain significant, the semiparametric tests appear to lose some power.

The results for the sign and magnitude of coefficients when estimated on EU MFP data differ substantially from the estimates using the full panel of TFP from the Penn World Tables in the main text. The coefficients on timber and forest cover swap signs; coefficients on timber land also become statistically significant at traditional levels in some specifications. To examine whether this difference in estimated coefficients is driven by heterogeneity across geographies within the main sample, we re-estimate our main specification on a subsample restricted to a subset of EU countries plus Norway. This “apples-to-apples” comparison re-runs the regression in Table 7 using the Penn World Tables TFP series as opposed to the Eurostat MFP series as an outcome variable. To ensure a fair comparison, we restrict the sample to the truncated time period of 2003-2019. During this period, the Pearson coefficient between the two measures of TFP growth is $\rho = 0.81$.

The comparison results are shown in Table 8. Inspection of the results when the EU27-plus-Norway outcome is re-estimated using PWT data yields qualitatively similar coefficient changes. The signs on forests and timber land do flip. However, while the tables pass a visual test in terms of appearing similar, the joint estimates of the coefficients are statistically different. When we estimate the specification in column (4) jointly for each outcome variable using a seemingly unrelated regression, a Wald test of the equality of all coefficients on natural capital between the two models rejects the null hypothesis of equality far beyond standard levels ($p < 0.001$).

USDA ERS Agricultural TFP We examine whether the renewable capital stocks we consider are correlated with a measure of TFP in the agriculture sector constructed by the USDA’s Economic

Research Service (Fuglie 2015). The TFP measure from Fuglie (2015) differs qualitatively from measures in the PWT in that it explicitly accounts for one of the natural capital inputs that we consider, agricultural land, as well as more specialized inputs such as the use of water and fertilizer as well as tillage. As such, we do not expect to see growth in agricultural land as a good predictor when regressing this measure of agricultural TFP growth on the natural capital stocks.

Table 9 shows results when equation 4 in the main text is re-estimated on a panel of agricultural TFP growth for 114 (163) countries between 1997 and 2016 depending on the availability of controls in the PWT. The coefficient on agricultural land area is statistically indistinguishable from zero, as expected given a different metric of agricultural land use is subtracted as an input when the outcome of $\Delta \ln(Ag. TFP_{j,t})$ is constructed (Fuglie 2015). Tests of the joint null hypothesis performed using asymptotic and semiparametric approaches are marginal, and coefficients on individual resources fluctuate substantially depending on the specification we consider.

Output Growth Finally, we consider GDP growth as an alternative proxy for TFP growth after residualizing the portion explained by growth in the inputs of produced capital and labor. We examine whether this correlation is present for growth (log changes) in two GDP series taken from the PWT: $RGDP^{NA}$ (real GDP in 2017 USD based on national accounts) and $CGDP^O$ (output-side real GDP at current purchasing power parity). Using GDP as an outcome does not per se test whether natural capital stocks are correlated with productivity, but a strong correlation between changes in GDP and natural capital suggests that the latter are important to production.

To ensure we are not simply capturing correlations between natural capital and known factor inputs (e.g., produced capital and labor), we include them as controls in all versions of Equation 6 from the main text that we estimate where output growth is the outcome variable. Table 10 shows the results for $RGDP^{NA}$ as the outcome variable and Table 11 shows the results for $CGDP^O$. After using OLS to partial out the portion of output and natural capital growth explained by factor inputs (i.e., control for g_K and g_L in the PWT), we can strongly reject the null hypothesis of no correlation between growth in natural capital stocks and growth in output as measured by $RGDP^{NA}$. Table 10 shows that the p -values of joint F tests using asymptotic and semiparametric methods are below standard significance levels in all specifications, giving further evidence that natural capital is an omitted input that is important to production. However, the results become less precise when we instead measure output growth using PPP-based output. Table 11 shows the results for the corresponding $CGDP^O$ series. We cannot reject the joint null hypothesis in any of the specifications we consider.

2.3 Asymptotic Properties of the Short Regression Estimator

Let j and t index observations from year $t \in \{1, \dots, T\}$ in country $j \in \{1, \dots, J\}$. Write $g_{S,j,t}$ as the vector of growth in country-level NK stocks, the vector $g_{X,j,t}$ as the growth in factor inputs, $\hat{m}_{j,t} \triangleq (\hat{\gamma}_i, \hat{g}_{i,j,t})$ as a vector of estimated output elasticities and growth rates of included inputs, $g_{TFP,j,t}$ as unobserved TFP growth, and $\mathbf{Z}_{j,t}$ as a vector containing a constant term and the set of

potential controls. Write the outcomes and covariates as the stacked random vector

$$\mathbf{g}_{j,t} = [g'_{S,j,t}, g'_{X,j,t}, \hat{m}'_{j,t}, g_{TFP,j,t}, \mathbf{Z}'_{j,t}]'$$

We assume the $\mathbf{g}_{j,t}$ are iid draws from a stationary underlying distribution with finite first and second moments. Our identification assumption relies on growth in the natural capital stocks—where $\tilde{g}_{S,j,t}$ is the vector of (potentially residualized) growth terms—being valid instruments for the omitted factors:

$$\mathbb{E} \left[\tilde{g}_{S,j,t} \left(g_{TFP,j,t} + \sum_{i \in \mathbf{I}} [\gamma_i g_{i,j,t} - \hat{\gamma}_i \hat{g}_{i,j,t}] \right) \right] = 0 \quad (8)$$

and that the moment condition

$$\mathbb{E}[\tilde{g}_{S,j,t} \tilde{g}'_{S,j,t}] \quad (9)$$

is finite and invertible. Standard results then confirm that the OLS estimator $\hat{\beta}_S$ is consistent and asymptotically normal.

2.3.1 Consistency

The OLS estimator for the vector of β_s is

$$\hat{\beta}_S = \left(\frac{1}{JT} \sum_{j=1}^J \sum_{t=1}^T \tilde{g}_{S,j,t} \tilde{g}'_{S,j,t} \right)^{-1} \left(\frac{1}{JT} \sum_{j=1}^J \sum_{t=1}^T \tilde{g}_{S,j,t} \tilde{g}_{A,j,t} \right) \quad (10)$$

where $\tilde{g}_{S,j,t}$ is a vector of (potentially residualized) natural capital stocks and $\tilde{g}_{A,j,t}$ is residualized TFP growth. Write the difference between the estimator and estimand as

$$\begin{aligned} \hat{\beta}_S - \beta_S = & \left(\frac{1}{JT} \sum_{j=1}^J \sum_{t=1}^T \tilde{g}_{S,j,t} \tilde{g}'_{S,j,t} \right)^{-1} \left(\frac{1}{JT} \sum_{j=1}^J \sum_{t=1}^T \tilde{g}_{S,j,t} \left[g_{TFP,j,t} + \sum_{x \in \mathbf{X} \setminus \mathbf{I}} \gamma_x g_{x,j,t} \right. \right. \\ & \left. \left. + \sum_{i \in \mathbf{I}} [\gamma_i g_{i,j,t} - \hat{\gamma}_i \hat{g}_{i,j,t}] \right] \right) - \beta_S \end{aligned} \quad (11)$$

When $\tilde{g}_{S,j,t}$ is uncorrelated with true TFP growth and measurement error

$$\mathbb{E} \left[\tilde{g}_{S,j,t} \left(g_{TFP,j,t} + \sum_{i \in \mathbf{I}} [\gamma_i g_{i,j,t} - \hat{\gamma}_i \hat{g}_{i,j,t}] \right) \right] = 0 \quad (12)$$

the first and third terms inside the brackets in (11) will p -lim out to zero by the weak law of large numbers. This leaves

$$\begin{aligned}
\lim_{T \rightarrow \infty} (\hat{\beta}_S - \beta_S) &= \lim_{T \rightarrow \infty} \left(\left(\frac{1}{JT} \sum_{j=1}^J \sum_{t=1}^T \tilde{g}_{S,j,t} \tilde{g}'_{S,j,t} \right)^{-1} \left(\frac{1}{JT} \sum_{j=1}^J \sum_{t=1}^T \tilde{g}_{S,j,t} \left[\sum_{x \in \mathbf{X} \setminus \mathbf{I}} \gamma_x g_{x,j,t} \right] \right) \right) - \beta_S \\
&= \mathbb{E}[\tilde{g}_{S,j,t} \tilde{g}'_{S,j,t}]^{-1} \mathbb{E}[\tilde{g}_{S,j,t} \tilde{g}_{A,j,t}] - \beta_S = 0
\end{aligned}$$

which implies that $\hat{\beta}_S \xrightarrow{p} \beta_S$ as long as the moment $\mathbb{E}[\tilde{g}_{S,j,t} \tilde{g}'_{S,j,t}]$ is finite and invertible as we've assumed.

2.3.2 Asymptotic normality

Starting from equation (11), we have

$$\begin{aligned}
\sqrt{JT} (\hat{\beta}_S - \beta_S) &= \left(\frac{1}{JT} \sum_{j=1}^J \sum_{t=1}^T \tilde{g}_{S,j,t} \tilde{g}'_{S,j,t} \right)^{-1} \left(\frac{1}{\sqrt{JT}} \sum_{j=1}^J \sum_{t=1}^T \tilde{g}_{S,j,t} \left[g_{TFP_{j,t}} + \sum_{x \in \mathbf{X} \setminus \mathbf{I}} \gamma_x g_{x,j,t} \right. \right. \\
&\quad \left. \left. + \sum_{i \in \mathbf{I}} [\gamma_i g_{i,j,t} - \hat{\gamma}_i \hat{g}_{i,j,t}] \right] \right) - \sqrt{JT} \beta_S
\end{aligned} \tag{13}$$

Assume a central limit theorem applies to the products between stock growths and underlying TFP growth along with measurement error such that

$$\frac{1}{\sqrt{JT}} \sum_{j=1}^J \sum_{t=1}^T \tilde{g}_{S,j,t} \left[g_{TFP_{j,t}} + \sum_{i \in \mathbf{I}} [\gamma_i g_{i,j,t} - \hat{\gamma}_i \hat{g}_{i,j,t}] \right] \xrightarrow{d} N(0, \Omega) \tag{14}$$

where

$$\Omega \triangleq \text{Var} \left(\tilde{g}_{S,j,t} \left[g_{TFP_{j,t}} + \sum_{i \in \mathbf{I}} [\gamma_i g_{i,j,t} - \hat{\gamma}_i \hat{g}_{i,j,t}] \right] \right)$$

Then, by Slutsky's Lemma, we have

$$\begin{aligned}
&\left(\frac{1}{JT} \sum_{j=1}^J \sum_{t=1}^T \tilde{g}_{S,j,t} \tilde{g}'_{S,j,t} \right)^{-1} \frac{1}{\sqrt{JT}} \sum_{j=1}^J \sum_{t=1}^T \tilde{g}_{S,j,t} \left[g_{TFP_{j,t}} \right. \\
&\quad \left. + \sum_{i \in \mathbf{I}} [\gamma_i g_{i,j,t} - \hat{\gamma}_i \hat{g}_{i,j,t}] \right] \xrightarrow{d} \mathbb{E}[\tilde{g}_{S,j,t} \tilde{g}'_{S,j,t}]^{-1} N(0, \Omega)
\end{aligned} \tag{15}$$

as the first term on the LHS above converges in probability to $\mathbb{E}[\tilde{g}_{S,j,t} \tilde{g}'_{S,j,t}]^{-1}$ by the law of large numbers. Then

$$\sqrt{JT} (\hat{\beta}_S - \beta_S) \xrightarrow{d} N \left(0, \mathbb{E}[\tilde{g}_{S,j,t} \tilde{g}'_{S,j,t}]^{-1} \Omega \mathbb{E}[\tilde{g}_{S,j,t} \tilde{g}'_{S,j,t}]^{-1} \right) \tag{16}$$

3 Producing Törnqvist Quantity Indices for Natural Capital Aggregates

The theory and simulations in the main text consider how the marginal addition of one natural capital stock to the growth accounting equation affects estimates in the two-stock world. While this in theory corresponds to the case of renewable and non-renewable resources, in reality there are multiple forms of natural capital under each of these classifications. Aggregating various non-renewable and renewable resources into single quantities requires the construction of index numbers, which are summary statistics for growth in the quantity of all renewable and non-renewable natural capital inputs used in production each year. This section describes the construction of Törnqvist quantity indices for renewable and non-renewable natural capital used to calibrate the simulations in Section 3 of the main text.

To aggregate the sets of renewable and non-renewable resources to single quantities, we form Törnqvist (Törnqvist et al. 1936) quantity indices for each resource. The Törnqvist quantity index for renewable (non-renewable) resources is a geometric mean over growth in each renewable (non-renewable) input that period, with each input weighted by its share of the total value of the full set of renewable (non-renewable) natural capital inputs. The process for constructing Törnqvist indices consists of three steps: (1) constructing quantities used in production and associated rents accruing to each resource in a given year, (2) computing value share weights, and (3) aggregating weighted growth into composite indices. It aligns closely with the procedure used by the World Bank when constructing the official *Changing Wealth of Nations* (CWON) wealth measures (World Bank 2025), but performs all calculations excluding resources where data on quantities extracted or rents are proprietary and not made public.

3.1 Natural Capital Use and Rent Data

Data on quantities of resources extracted (for the case of non-renewables) and levels of stocks (for the case of renewables) are drawn directly from the World Bank CWON dataset (World Bank 2025). Table 12 shows the 13 non-renewable natural capital stocks that are inputs to the non-renewable Törnqvist quantity index. Non-renewables consist of the three main fossil fuels that comprise roughly four-fifths of global energy use (i.e., coal, oil, and natural gas), most of the common ferrous and base metals, and a few precious metals that are used in production. Table 13 shows the six renewable natural capital stocks that are inputs to the renewable Törnqvist quantity index. These resources align directly with those used as independent variables in the regression analysis listed in Table 2.

3.2 Data Construction

For each non-renewable resource, quantities and rents (i.e., income attributable to that resource in that year) are taken directly from the CWON dataset at the country-by-year level. Quantities of

each resource r extracted in country j in year t are denoted as

$$Q_{j,t}^{(r)}$$

and are based on physical quantities (e.g., barrels of oil produced or tons of minerals mined). This approach considers only the use of non-renewable resources as inputs to extractive industries, where resources in their raw form are transformed into commodities sold as intermediate or final goods. This measure of non-renewable inputs differs substantially from the case where they are intermediate goods in production (e.g., crude oil as an input to oil refining), which are excluded when calculating GDP to avoid double counting. The associated rents are denoted

$$R_{j,t}^{(r)}$$

and are calculated by the World Bank based on quantities extracted multiplied by local or global commodity market prices (World Bank 2024).

For the case of renewables, quantities are constructed in the same manner, but based on physical measurements of stocks, rather than the flow of services provided. As discussed in the main text, this requires assuming that the services renewable natural capital stocks provide are directly proportional to the physical quantities used in production. Specifically, we require that services provided per unit of stock of a given resource are the same across countries and years. This allows us to set the percent changes in renewable service flows each year equal to the percent changes in stocks, an object that is measured in the CWON data. We assign annual rents for renewables based on the flow value imputed to each service in the CWON data. These values are based on various studies that use direct or non-market valuation methods to assess the annual benefits from physical quantities of natural capital stocks. For example, the value of farmland is assigned based on the output of the agriculture it facilitates, while mangroves are valued based on the flood protection they provide (World Bank 2024). All resource datasets are merged into a single country-by-year panel covering non-renewable (e.g., fossil fuels, minerals) and renewable (e.g., forests, agricultural land, hydropower) quantities and rents.

3.3 Gross Growth Rates

For each series from the CWON giving the quantity of a given resource used in country j , the gross growth rate is computed as:

$$\hat{Q}_{j,t}^{(r)} = \frac{Q_{j,t}^{(r)}}{Q_{j,t-1}^{(r)}}$$

To avoid leverage from outliers driven by either anomalous growth or administrative errors in data entry, growth values are top-coded at a ratio of 2 (i.e., extraction is assumed to not grow at a rate

larger than 100% per annum):

$$\hat{Q}_{j,t}^{(r)} = \min(\hat{Q}_{j,t}^{(r)}, 2)$$

Between 0 and 265 values for gross growth rates are winsorized for each resource. Observations affected by missing data or zero base-period values are normalized to one. After these adjustments, the dataset has 11,067 entries, a balanced panel of 217 geographic entities between 1970 and 2020. A majority (between 6,480 and 9,942) of entries are initially missing for all resources and replaced with a gross growth rate of one indicating no change.

3.4 Törnqvist Weights

Non-renewable resources: Total rents paid to non-renewable natural capital each period in country i are calculated as

$$R_{j,t}^{NR} = \sum_{r \in \mathcal{N}} R_{j,t}^{(r)}$$

where $\mathcal{N}$ is the set of non-renewable resources given in Table 12. The value share for each resource (its share of total rents across all non-renewable resources that period) is calculated using:

$$w_{j,t}^{(r)} = \frac{R_{j,t}^{(r)}}{R_{j,t}^{NR}}$$

These value shares are used to construct period- t Törnqvist weights based on an equally-weighted average of rent shares in period t and $t - 1$

$$\tilde{w}_{j,t}^{(r)} = \frac{1}{2} (w_{j,t}^{(r)} + w_{j,t-1}^{(r)})$$

For non-renewable resources, we are able to construct rent weights for 6,064 observations at the country level, an unbalanced panel across all 217 countries.

Renewable resources: Total renewable rent each year is defined analogously as

$$R_{j,t}^R = \sum_{k \in \mathcal{R}} R_{j,t}^{(k)}$$

and weights in each year are computed using:

$$w_{j,t}^{(k)} = \frac{R_{j,t}^{(k)}}{R_{j,t}^R}$$

For renewable resources, we are able to construct rent weights for 3,684 observations at the country level, an unbalanced panel across 149 countries between 1995 and 2020. Because this leaves us with a substantial number of observations for which quantities are observed while rent weights are

missing, each resource is assigned the average value of its weights across all years where data are available $\mathcal{T}_j$:

$$\bar{w}_j^{(k)} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} w_{j,t}^{(k)}$$

when constructing the indices for renewable inputs for each country.

3.5 Törnqvist Indices

Non-renewable resources: For each resource r in the set of non-renewables $\mathcal{N}$, individual gross changes in quantities are then exponentiated:

$$Q_{j,t}^{(r)*} = \left(\hat{Q}_{j,t}^{(r)} \right)^{\bar{w}_{j,t}^{(r)}}$$

The composite Törnqvist quantity index for non-renewable inputs $\mathcal{Q}_{j,t}^N$ is the product across resources of all exponentiated terms.

$$\mathcal{Q}_{j,t}^N = \prod_{r \in \mathcal{N}} Q_{j,t}^{(r)*}$$

Growth in the non-renewable index is then defined as

$$g_{j,t}^{NR} \triangleq \mathcal{Q}_{j,t}^N - 1$$

when used for calculating in-sample means, standard deviations, and correlations for non-renewable resource use growth at the national level.

Renewable resources: The renewable Törnqvist index $\mathcal{Q}_{j,t}^R$ is constructed in an analogous manner. First, gross quantity changes for all renewable resources k are exponentiated using the time-invariant rent share weights

$$Q_{j,t}^{(k)*} = \left(Q_{j,t}^{(k)} \right)^{\bar{w}_i^{(k)}}$$

and the index is computed as

$$\mathcal{Q}_{j,t}^R = \prod_{k \in \mathcal{R}} Q_{j,t}^{(k)*}$$

Growth in the renewable natural capital Törnqvist quantity index is then set as

$$g_{j,t}^R \triangleq \mathcal{Q}_{j,t}^R - 1$$

when used for calculating correlations at the national level. Finally, $g_{j,t}^R$ and $g_{j,t}^{NR}$ are topcoded at 1, ruling out over 100% growth in the quantity indices. This affects 86 instances of the renewables

index and zero instances of the non-renewables index.

3.6 Regression Analysis

The Törnqvist indices for renewable and non-renewable capital inputs are intended to capture growth in the factor inputs stemming from renewable and non-renewable natural capital at the national level. We suspect that these inputs, especially non-renewable hydrocarbons, should explain some portion of output growth. To ensure growth in our indices for the use of non-renewable and renewable capital is positively associated with output, we run the following regression

$$\Delta \ln(y_{j,t}) = \gamma_1 g_{j,t}^{NR} + \gamma_2 g_{j,t}^R + \theta' \mathbf{Z}_{j,t} + \delta_j + \zeta_t + \varepsilon_{j,t} \quad (17)$$

where $y_{j,t}$ is the logarithm of real GDP at constant national prices ($RGDP^{NA}$) taken from the PWT intended for use in measuring GDP growth (Feenstra, Inklaar, and Timmer 2015). The set $\mathbf{Z} = \{K, L, HC, LS\}$ contains controls for growth in produced capital, employment, human capital, and the labor share in each country each year. Terms (δ_j, ζ_t) are again two-way fixed effects, and $\varepsilon_{j,t}$ are idiosyncratic error terms. The coefficients γ_1 and γ_2 —the output elasticities for our indices of non-renewable and renewable capital—are our estimands of interest and the feasible analogs of the theoretical counterparts in the simulation section.

Table 14 shows the resulting estimates of γ_1, γ_2 from five variants of the specification in equation 17 when estimated on an unbalanced panel of either 104 or 134 countries between 1972 and 2019 (the number of observations and covered countries depends on the availability of controls). Parenthesis show heteroskedasticity-robust standard errors. Consistent with the expectation that growth in the use of natural capital would be associated with output growth, estimates of output elasticities γ are positive across all specifications for growth in both input indices. Point estimates for the output elasticities of non-renewable and renewable natural capital services hover around 0.02 and 0.06, respectively. For non-renewables, the estimates are highly significant across all specifications ($p < 0.01$) and comparable in magnitude to the output elasticity of energy (which is typically composed mainly of fossil fuels) found in the literature using Cobb-Douglas specifications for aggregate output (see Golosov et al. 2014; Casey 2024). The coefficients on renewables, while larger in magnitude, are slightly less precise but still significant at standard levels. Given the results in Table 14 for $\hat{\gamma}_2$, we assign γ_2 as 0.05 when calibrating the simulations described in Section 4.

4 Country-Level Simulation Exercise

We now describe the simulation exercise underpinning the results in Section 3 of the main text. Section 4.1 starts by giving a brief rehash of the simulation exercise described in the main text that gives a more complete explanation of the calibration procedure.

4.1 Setup

Data Generating Process for Output Starting with the model in the main text, we assume output growth in each country comprises growth in productivity, labor, produced capital, and two forms of natural capital inputs (non-renewables N_1 and renewables N_2). Joint observations of growth rates for all determinants of output growth are assumed to be independent and identically distributed draws of vector $\mathbf{g}_{j,t}$ from a fixed distribution for each country with mean $\boldsymbol{\mu}_j$ and variance Σ_j . Output growth in each simulated country j is generated by

$$g_{Y,j,t} = \mathbf{g}_{j,t}' \boldsymbol{\gamma}_j \quad (18)$$

where $\boldsymbol{\gamma}_j \triangleq [1, \gamma_K, \gamma_L, \gamma_1, \gamma_2]'$ is the vector of output elasticities for that country.

Structural Parameters We set $\gamma_2 = 0.05$ based on the analysis in Section 3.6. In turn, to induce an overall output elasticity of 0.1 for natural capital (following Arrow et al. 2004), which we view as a lower bound, we set γ_1 as 0.05.⁶ This ad hoc choice is slightly higher than our point estimates for this elasticity based on the reduced-form analysis in Section 3.6, which hover between 0.01 and 0.03. However, given existing estimates for the average factor shares associated with natural resources excluding urban land (5.44 percent from Monge-Naranjo, Sánchez, and Santaaulalia-Llopis (2019) and between 4.8 and 9.1 in Manning and Barbier 2025), the output elasticity of energy inputs historically (Goloso et al. 2014; Casey 2024) as well as fossil fuel’s share of historical factor income (Hassler, Krusell, and Olovsson 2021) in the literature, we view this upward adjustment as reasonably capturing the output elasticity of the combined composite of all non-renewable resources.⁷

Procedure for Estimating TFP Growth We assume an econometrician observes a subset of growth in factor inputs—those for K , L , and N_1 —along with growth in output without error for 25 periods (years) in each country. We select a sample size of 25 periods because the CWON data on natural capital cover this many years in each country. She deploys the model for estimating TFP growth while omitting non-renewable resources (model 1) by estimating:

6. This is a comfortable lower bound given existing evidence on the share of output that accrues to all forms of natural capital; Monge-Naranjo, Sánchez, and Santaaulalia-Llopis (2019) find a median (mean) factor share of 14.7 (17.7) percent for natural resources including urban land for a cross section of 79 countries in version 8.0 of the Penn World Tables.

7. Note we need not make assumptions on γ_L and γ_K in order to solve for the bias or mean squared error of the intercept term.

$$g_Y = \hat{\gamma}_A + \hat{\gamma}_K g_K + \hat{\gamma}_L g_L + \epsilon \quad (19)$$

on her time series of data for growth in $\{K, L\}$ in a given country using ordinary least squares (OLS). Her estimator for average TFP growth under model 1 is the intercept term $\hat{\gamma}_A$. She estimates the analog for the case including non-renewable resources (model 2)

$$g_Y = \tilde{\gamma}_A + \tilde{\gamma}_K g_K + \tilde{\gamma}_1 g_{N_1} + \tilde{\gamma}_L g_L + \nu \quad (20)$$

on time series for growth in $\{K, L, N_1\}$. The OLS-based estimator for average TFP growth under model 2 is $\tilde{\gamma}_A$.

Parameters Governing Growth in Inputs We assume at the country level that growth in TFP (g_{TFP}) is uncorrelated with growth in other factors. This restriction can be relaxed to allow arbitrary correlations between growth in factor inputs and true TFP growth, but we elect to set these correlations as zero in all countries as we do not have strong priors for what these unobservable correlations should be. The simulation procedure would remain virtually unchanged in this case apart from assigning alternative values to those off-diagonal terms in Σ_j for each country.

Calculating the bias and mean squared error for the TFP estimators from models 1 and 2 requires knowing the mean (μ_j) and variance (Σ_j) governing the joint distribution of growth terms g_j in each country.⁸ A description of each moment, as well as the sources we use to calibrate it, are given in Table 15. We set the parameters governing average growth in the use of natural capital equal to the sample means for the growth in renewable and non-renewable Törnqvist indices constructed in Section 3 between 1996 and 2019 (Table 15 rows one and two). Rows three, four, and five, governing expected growth in produced capital services, labor, and TFP, are set equal to their sample analogs in the Penn World Tables (Feenstra, Inklaar, and Timmer 2015) over the same 1996-2019 period. Finally, higher-order moments are set based on the empirical covariances between growth in the Törnqvist non-renewable index and growth in other inputs we take from the PWT during the 1996-2019 period. Given the limited overlap in data between the CWON and PWT data, we are ultimately able to calculate sample analogs of μ and Σ for 100 countries, which we substitute directly into the formulas for bias and RMSE below.

4.2 Bias in Estimated TFP

We now turn to calculating the bias and mean squared error for each estimator at the country level. As shown in Section 6.1, when g_{TFP} is independent and identically distributed, the bias for the estimate of average TFP growth from model 1 is

8. Technically, we need not know $\mathbb{E}[g_{TFP}]$ to calculate the bias or expected MSE. However, we do use this to calculate a ballpark ratio of bias reduction to true underlying growth, so it is treated as needed here.

$$\mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}]] = \gamma_1 \mathbb{E}[g_{N_1} - \kappa_{N_1, \tilde{K}} g_K - \kappa_{N_1, \tilde{L}} g_L] + \gamma_2 \mathbb{E}[g_{N_2} - \kappa_{N_2, \tilde{K}} g_K - \kappa_{N_2, \tilde{L}} g_L] \quad (21)$$

where the $\kappa_{N_j, \cdot}$ coefficients are generated by the projections of g_{N_1} and g_{N_2} onto the vector of included factor growths and a constant, $\hat{\mathbf{g}} = [1, g_K, g_L]'$:

$$\begin{aligned} \kappa_{N_1} &= \mathbb{E}[\hat{\mathbf{g}} \hat{\mathbf{g}}']^{-1} \mathbb{E}[\hat{\mathbf{g}} g_{N_1}] \\ \kappa_{N_2} &= \mathbb{E}[\hat{\mathbf{g}} \hat{\mathbf{g}}']^{-1} \mathbb{E}[\hat{\mathbf{g}} g_{N_2}] \end{aligned}$$

For the case of Model 2 where only N_2 is omitted, the bias is

$$\mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}]] = \gamma_2 \mathbb{E}[g_{N_2} - \lambda_{N_2, \tilde{K}} g_K - \lambda_{N_2, \tilde{L}} g_L - \lambda_{N_2, \tilde{N}_1} g_{N_1}] \quad (22)$$

where the $\lambda_{N_2, \cdot}$ coefficients are generated by the projection of g_{N_2} onto $\tilde{\mathbf{g}} = [1, g_K, g_L, g_{N_1}]'$:

$$\boldsymbol{\lambda} = \mathbb{E}[\tilde{\mathbf{g}} \tilde{\mathbf{g}}']^{-1} \mathbb{E}[\tilde{\mathbf{g}} g_{N_2}]$$

4.3 Expected Improvements in Root Mean Squared Error

Under the assumption of homoskedastic errors (constant variance in underlying TFP growth across observations within a given country) and independently sampled growth terms, Section 6.2 shows the estimator $\hat{\gamma}$ in (19) has conditional variance

$$\text{Var}(\hat{\gamma}|\hat{\mathbf{g}}) = \left(\sigma_{TFP}^2 + [\gamma_1, \gamma_2] \Sigma_{1,2|\mathbf{g}} \begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix} \right) (\hat{\mathbf{g}} \hat{\mathbf{g}}')^{-1}$$

where here in a slight abuse of notation, $\hat{\mathbf{g}} = [\mathbf{1}, g_K, g_L]'$ is the 3×25 matrix of observed growth terms (and a constant) included under model 1 and $\Sigma_{1,2}$ is the (conditional) covariance matrix for g_{N_1} and g_{N_2} . The conditional variance of the intercept term is then

$$\text{Var}(\hat{\gamma}_A|\hat{\mathbf{g}}) = \frac{\sigma_{TFP}^2 + \gamma_1^2 \sigma_1^2 + \gamma_2^2 \sigma_2^2 + 2\gamma_1 \gamma_2 \sigma_{1,2}}{25}$$

where σ_i^2 and $\sigma_{1,2}$ are the (conditional) variances and covariance of growth in the natural capital stocks.

Section 6.2 shows the estimator of TFP from model 2 has variance

$$\text{Var}(\tilde{\gamma}_A|\tilde{\mathbf{g}}) = \frac{\sigma_{TFP}^2 + \gamma_2^2 \sigma_2^2}{25}$$

where $\tilde{\mathbf{g}} = [1, g_K, g_L, g_{N_1}]$ is the matrix of regressors under Model 2.

We calculate the expected change in the root mean squared error term— Δ RMSE—as

$$\Delta \text{RMSE} \triangleq \sqrt{\mathbb{E}[(\hat{\gamma}_A - \mathbb{E}[g_{TFP}])^2 | \hat{\mathbf{g}}]} - \sqrt{\mathbb{E}[(\tilde{\gamma}_A - \mathbb{E}[g_{TFP}])^2 | \tilde{\mathbf{g}}]} \quad (23)$$

which can be decomposed as

$$\Delta \text{RMSE} \equiv \sqrt{\text{Var}(\hat{\gamma}_A | \hat{\mathbf{g}}) + \mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}] | \hat{\mathbf{g}}]^2} - \sqrt{\text{Var}(\tilde{\gamma}_A | \tilde{\mathbf{g}}) + \mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}] | \tilde{\mathbf{g}}]^2} \quad (24)$$

as $\mathbb{E}[g_{TFP}]$ is a constant and growth in TFP is assumed iid. The (squared) bias terms δ and ε are defined in Section 4.2 and become biases conditional on the observed portion of $\mathbf{g}$ when placed inside the operator $\mathbb{E}[\cdot | \mathbf{g}]$.

4.4 Additional Results from the Simulations

4.4.1 Decomposition of RMSE Improvement

As referenced in Section 3 of the main text, the vast majority of RMSE in estimated TFP growth under models 1 and 2 stems from the bias terms. Figure 2 gives a histogram of the share of RMSE attributable to the bias terms

$$\text{Share 1} = \frac{\mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}] | \hat{\mathbf{g}}]^2}{\text{Var}(\hat{\gamma}_A | \hat{\mathbf{g}}) + \mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}] | \hat{\mathbf{g}}]^2}$$

for the model omitting both NK stocks and

$$\text{Share 2} = \frac{\mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}] | \tilde{\mathbf{g}}]^2}{\text{Var}(\tilde{\gamma}_A | \tilde{\mathbf{g}}) + \mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}] | \tilde{\mathbf{g}}]^2}$$

for the model including non-renewables.

Figure 3 gives a sense of the reduction in (squared) bias and the corresponding (lack of) reduction in variance when going from the estimator $\hat{\gamma}_A$ to $\tilde{\gamma}_A$.

4.4.2 Constructing the Country-Level Overlay on Figure 1

Table 16 shows data for the set of 100 individual countries mapped in Figure 1a (and plotted in Figures 1b and 2) in the main text. The two columns labeled Bias and RMSE Reduction denote the effect of adopting an estimator including non-renewable resources on the accuracy of the TFP estimate at the country level in terms of unconditional bias reductions and conditional on 25 years of observed data.

4.4.3 Reduced-Form Decomposition of RMSE Improvement Across Individual Moments of Input Growth

Table 17 shows the regression coefficients when ΔRMSE is projected onto the elements of the sample versions of μ_j and Σ from Section 4.3. The test statistic for the joint null hypothesis that the coefficients on $\rho(g_{N_1}, g_K)$ and $\rho(g_{N_1}, g_{N_2})$ in Table 17 are both zero is 0.039. When the regression is restricted to a bivariate one containing only $\rho(g_{N_1}, g_K)$ and $\rho(g_{N_1}, g_{N_2})$, the test statistic for the joint null falls below 0.01. $\rho(g_{N_1}, g_K)$ and $\rho(g_{N_1}, g_{N_2})$ are also retained as regressors when applying a 10-fold cross-validation LASSO procedure. Figure 4 gives a visual version of each regression coefficient by plotting residualized values ΔRMSE against the 12 different sample moments in each country. The slopes of the dashed black lines in Panels a-l correspond to the regression coefficients in Table 17.

4.5 Sensitivity Analysis of Simulation Results

The applicability of our simulation results in Section 3 of the main text relies on our choices of parameters governing growth in output and factor inputs accurately reflecting reality. While our choice of parameters is based on sample moments (i.e., Table 15), our own econometric analysis (i.e., Table 14), or estimates from prior literature (e.g., Arrow et al. 2004; Hassler, Krusell, and Olovsson 2021), they play a critical role in determining whether including non-renewables improves estimates of TFP. In this section we test the sensitivity of our simulation results along two fronts: choices of output elasticities for natural capital (γ_1 and γ_2) and the sample periods for estimating the growth rates of inputs at the country level (μ_j and Σ_j).

4.5.1 Choice of Output Elasticities

This subsection tests the sensitivity of our results to choices of elasticities γ_1 and γ_2 . To do this, we run a Monte Carlo analysis over draws of random pairs of alternative output elasticities drawn from iid uniform distributions ($\mathcal{U}[0, 0.1] \times \mathcal{U}[0, 0.1]$) where each pair is set as alternative output elasticities for non-renewables and renewables for a random country in our sample. We store the resulting ΔRMSE for country j under these alternative assumptions for γ_1, γ_2 holding all other relevant moments in (μ_j and Σ_j) unchanged. We repeat this process 10,000,000 times (roughly 100,000 times per country).

Figure 5 shows a histogram of the results from the Monte Carlo exercise. Each individual MC draw produces an alternative ΔRMSE_j for a given country, given a random pair of output elasticities for natural capital, γ_1, γ_2 , drawn from $\mathcal{U}[0, 0.1]$. The median (mean) reduction in RMSE at the country level across our 10,000,000 alternative draws falls to 2 (28) basis points. The distribution remains quite dispersed, displaying a standard deviation of 80 basis points. Unlike with our main analysis, a substantial share (31 percent) of country-level outcomes are negative under random alternative elasticities (as shown by the substantial mass of the histogram to the left of the vertical dashed line at zero).

4.5.2 Choice of Sample Period

To test the sensitivity of our results to the choice of sample period for constructing the sample analogs of population moments μ_j and Σ_j , we split our 1996-2019 sample into early (1996-2008) and late (2009-2019) subsample periods and repeat the simulation exercise. This reduces the set of countries from 100 to 93 and 98, respectively, in each sample period due to data limitations.

For the early period (1996-2008), the median (mean) reduction in RMSE rises to 16 (84) basis points. For the late period, the median and mean reductions in RMSE rise further to 22 and 98 basis points. The resulting changes in RMSE are also substantially more dispersed, as the sample versions of Σ become fairly large for the small number of periods covered by each subsample. The standard deviation of ΔRMSE rises to 155 and 200 basis points for the early and late periods, respectively. Our qualitative result in the main text, that including non-renewable natural capital helps on the margin, remains relatively unchanged. Ten countries display increases in RMSE when going from model 1 to model 2 when estimated on the early period subsample, while four display increases when using the late period subsample.

Figures 6 and 7 reconstruct alternative versions of Figure 2 in the main text for each subsample. The qualitative nature of our results—that including natural capital on the margin substantially improves estimates of TFP growth for most countries—remains true for both subsample periods.

5 Hypothetical Fiscal Forecast Errors

The \$300 billion figure referenced in Section 3 of the main text is constructed as follows. We assume that initial output in the United States is \$30 trillion nominal USD in 2025 and grows at a rate of five percent each year, roughly in line with historical averages for the U.S. since 1990 ([U.S. Bureau of Economic Analysis](#)). We assume one percentage point of this growth is caused by growth in TFP, while inflation and growth in other factor inputs account for the remaining four points. This leads to 10-year cumulative output between 2026 and 2035 of \$396.2 trillion dollars. If an econometrician instead overestimates that TFP will grow at a rate of 1.07 percent a year (a 7 basis point forecast error in each year) over the next ten years and while correctly accounting for other drivers of four points' worth of nominal growth accounting each year, she will forecast an average growth rate of 5.07 percent. This will generate a forecast of \$397.7 trillion in cumulative output between 2026 and 2035. The difference in forecast output is \$1.5 trillion. Assuming a ratio of tax receipts to GDP of roughly 20% (slightly above historical averages in the United States, see [U.S. Office of Management and Budget and Federal Reserve Bank of St. Louis](#)) this translates to an overshoot of (cumulative) revenue forecasts of \$300 billion.

6 Proofs

6.1 Biases of OLS-Based TFP Estimators

We present the bias calculations here for the general case where the set of natural capital stocks has cardinality $J \geq 2$. The bias in expected TFP growth when factor shares are estimated using the specification in (19) with OLS

$$\begin{aligned}\mathbb{E}[\hat{g}_A - \mathbb{E}[g_{TFP}]] &= \mathbb{E}[g_y] - \mathbb{E}[g_y] + [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - \hat{\gamma}_L]\mathbb{E}[g_L] + \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j}] \\ &= - \left[\sum_{j=1}^J \gamma_j \kappa_{N_j, \tilde{K}} + \kappa_{A, \tilde{K}} \right] \mathbb{E}[g_K] - \left[\sum_{j=1}^J \gamma_j \kappa_{N_j, \tilde{L}} + \kappa_{A, \tilde{L}} \right] \mathbb{E}[g_L] + \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j}] \\ &= \sum_{j=1}^J \gamma_j \mathbb{E}[g_{N_j} - \kappa_{N_j, \tilde{K}} g_K - \kappa_{N_j, \tilde{L}} g_L] - \kappa_{A, \tilde{K}} \mathbb{E}[g_K] - \kappa_{A, \tilde{L}} \mathbb{E}[g_L]\end{aligned}$$

where the κ_{N_j} coefficients are generated by linear projections of g_{N_j} onto the vector $[g_K, g_L]'$. The bias generated by the OLS specification in (20) follows in a similar manner

$$\begin{aligned}\mathbb{E}[\tilde{g}_A - \mathbb{E}[g_{TFP}]] &= - \left[\sum_{j=2}^J \gamma_j \lambda_{N_j, \tilde{K}} + \lambda_{A, \tilde{K}} \right] \mathbb{E}[g_K] - \left[\sum_{j=2}^J \gamma_j \lambda_{N_j, \tilde{L}} + \lambda_{A, \tilde{L}} \right] \mathbb{E}[g_L] \\ &\quad - \left[\sum_{j=2}^J \gamma_j \lambda_{N_j, \tilde{N}_1} + \lambda_{A, \tilde{N}_1} \right] \mathbb{E}[g_{N_1}] + \sum_{j=2}^J \gamma_j \mathbb{E}[g_{N_j}] \\ &= \sum_{j=2}^J \gamma_j \mathbb{E}[g_{N_j} - \lambda_{N_j, \tilde{K}} g_K - \lambda_{N_j, \tilde{L}} g_L - \lambda_{N_j, \tilde{N}_1} g_{N_1}] \\ &\quad - \lambda_{A, \tilde{K}} \mathbb{E}[g_K] - \lambda_{A, \tilde{L}} \mathbb{E}[g_L] - \lambda_{A, \tilde{N}_1} \mathbb{E}[g_{N_1}]\end{aligned}$$

where λ_{N_2} are coefficients from a projection of g_{N_2} onto $[g_K, g_L, g_{N_1}]$.

6.2 Variance of the OLS-Based TFP Estimators

Model 1 Split the true data-generating process from (18) into vectors of included and excluded regressors

$$y = Z\gamma_Z + X_o\gamma_o + \varepsilon \tag{25}$$

where

$$Z = [\mathbf{1}, g_K, g_L], \quad X_o = [g_{N_1}, g_{N_2}],$$

and

$$\gamma_Z = (\mathbb{E}[g_{TFP}], \gamma_K, \gamma_L)', \quad \gamma_o = (\gamma_1, \gamma_2)'$$

We assume here that the error term ε —the portion of output growth unexplained by input growth each period—is governed by residual variation in TFP growth such that

$$\mathbb{E}[\varepsilon | \mathbf{g}] = 0 \quad \text{Var}(\varepsilon) = \sigma_{TFP}^2$$

The OLS estimator from regressing y on Z alone is

$$\hat{\gamma} = (Z'Z)^{-1}Z'y. \quad (26)$$

Substituting the true model,

$$\hat{\gamma} = (Z'Z)^{-1}Z'(Z\gamma_Z + X_o\gamma_o + \varepsilon) \quad (27)$$

$$= \gamma_Z + (Z'Z)^{-1}Z'X_o\gamma_o + (Z'Z)^{-1}Z'\varepsilon\gamma_o \quad (28)$$

Define the combined error term as

$$u^{(1)} \triangleq X_o\gamma_o + \varepsilon \quad (29)$$

The variance of $\hat{\gamma}$ conditional on Z is given by the traditional sandwich estimand

$$\text{Var}(\hat{\gamma} | Z) = (Z'Z)^{-1}Z'\Omega_1Z(Z'Z)^{-1} \quad (30)$$

where

$$\Omega_1 \triangleq \text{Var}(u^{(1)} | Z)$$

is a covariance matrix of size $n = 25$. Under the assumption of homoskedastic errors and independent sampling we can write the covariance matrix Ω_1 as

$$\Omega_1 = \tau_1^2 I_n$$

where I_n is the identity matrix of size equal to the sample size n and τ_1^2 is a function of the variance of TFP growth σ_{TFP}^2 along with the variance of the omitted terms

$$\tau_1^2 = \sigma_{TFP}^2 + \text{Var}(\gamma_1 g_{N_1} + \gamma_2 g_{N_2} | Z) \quad (31)$$

Letting

$$\Sigma_{12|Z} \triangleq \text{Var}(X_o | Z)$$

we can write the variance term as

$$\text{Var}(\gamma_1 g_{N_1} + \gamma_2 g_{N_2} \mid Z) = \gamma_o' \Sigma_{12|Z} \gamma_o \quad (32)$$

This gives the result in Section 4.3:

$$\text{Var}(\hat{\gamma} \mid Z) = \left(\sigma_{TFP}^2 + \gamma_o' \Sigma_{12 \cdot Z} \gamma_o \right) (Z'Z)^{-1} \quad (33)$$

Model 2 Rewrite the true data-generating process from (DGP) as

$$y = W\gamma_W + g_{N_2}\gamma_2 + \varepsilon \quad (34)$$

where, similar to the procedure for Model 1, we assume

$$\mathbb{E}[\varepsilon|\mathbf{g}] = 0 \quad \text{Var}(\varepsilon) = \sigma_{TFP}^2 \quad W = [\mathbf{1}, g_K, g_L, g_{N_1}], \quad \gamma_W = (\mathbb{E}[g_{TFP}], \gamma_K, \gamma_L, \gamma_1)'$$

The OLS estimand from regressing y on W is

$$\tilde{\gamma} = (W'W)^{-1}W'y. \quad (35)$$

Substituting gives

$$\tilde{\gamma} = (W'W)^{-1}W'(W\gamma_W + g_{N_2}\gamma_2) \quad (36)$$

$$= \gamma_W + (W'W)^{-1}W'g_{N_2}\gamma_2 + (W'W)^{-1}W'\varepsilon \quad (37)$$

Define the error for model 2 as

$$u^{(2)} \triangleq \gamma_2 g_{N_2} + \varepsilon \quad (38)$$

Then the expected variance conditional on W is

$$\text{Var}(\tilde{\gamma} \mid W) = (W'W)^{-1}W'\Omega_2 W(W'W)^{-1} \quad (39)$$

where

$$\Omega_2 \triangleq \text{Var}(u^{(2)} \mid W).$$

Under the assumption of homoskedastic errors and independent sampling we can write the covariance matrix Ω_2 as

$$\Omega_2 = \tau_2^2 I_n,$$

where (analogous to above)

$$\tau_2^2 = \sigma_{TFP}^2 + \gamma_2^2 \text{Var}(g_{N_2} \mid W). \quad (40)$$

Substituting gives

$$\text{Var}(\tilde{\gamma} \mid W) = \left(\sigma^2 + \gamma_2^2 \text{Var}(g_{N_2} \mid W) \right) (W'W)^{-1} \quad (41)$$

as written in Section [4.3](#).

7 Tables

Table 1: Covered Countries

Angola	Argentina	Armenia
Australia	Austria	Bahrain
Barbados	Belgium	Benin
Bolivia	Botswana	Brazil
Bulgaria	Burkina Faso	Burundi
Cameroon	Canada	Central African Republic
Chile	China	Colombia
Costa Rica	Côte d'Ivoire	Croatia
Cyprus	Czech Republic	Denmark
Dominican Republic	Ecuador	Egypt
Estonia	Eswatini	Fiji
Finland	France	Gabon
Germany	Greece	Guatemala
Honduras	Hong Kong	Hungary
Iceland	India	Indonesia
Iran	Iraq	Ireland
Israel	Italy	Jamaica
Japan	Jordan	Kazakhstan
Kenya	South Korea	Kuwait
Kyrgyzstan	Laos	Latvia
Lesotho	Lithuania	Luxembourg
Macao	Malaysia	Malta
Mauritania	Mauritius	Mexico
Moldova	Mongolia	Morocco
Mozambique	Namibia	Netherlands
New Zealand	Nicaragua	Niger
Nigeria	Norway	Panama
Paraguay	Peru	Philippines
Poland	Portugal	Qatar
Romania	Russian Federation	Rwanda
Saudi Arabia	Senegal	Serbia
Sierra Leone	Singapore	Slovakia
Slovenia	South Africa	Spain
Sri Lanka	Sudan	Sweden
Switzerland	Tajikistan	Tanzania
Thailand	Togo	Trinidad and Tobago
Tunisia	Türkiye	Ukraine
United Kingdom	United States	Uruguay
Venezuela	Zambia	Zimbabwe

Table 2: Renewable resources used to estimate equation 4 in the main text

Resource	Unit	% Obs. Missing	# Countries
Productive timber forest area	Hectares	1.25%	114
Agricultural land area	Square Kilometers	2.56%	116
Non-timber forest cover	Square Kilometers	3.95%	113
Mangrove ecosystem area	Hectares	63.25%	43
Fishery Biomass	Metric tons	23.08%	90
Hydroelectric Power Generation	Gigawatt-hours	27.07%	86

Table 3: Additional Wald Tests

Model	Estimator	Wald p -value	Wild Bootstrap p -value
BCH	PDS	0.002	0.004
BCH	LASSO	0.000	0.007
BCH	post-LASSO	0.001	0.007
Driscoll-Kraay	TWFE	0.000	0.000
Driscoll-Kraay	TWFE + Controls	0.000	0.000
Driscoll-Kraay	TWFE + Controls + Trends	0.000	0.004

Table 4: TFP Growth vs. Natural Capital Growth, Excluding Land

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Forest Cover	0.068 (0.061)	0.027 (0.062)	0.047 (0.068)	0.067 (0.075)	0.086 (0.043)	-0.013 (0.070)
Mangroves	0.138 (0.240)	0.085 (0.237)	0.348 (0.298)	0.317 (0.322)	0.303 (0.366)	-0.096 (0.428)
Biomass	0.114 (0.098)	0.031 (0.078)	0.092 (0.107)	0.070 (0.101)	0.039 (0.134)	-0.109 (0.107)
Hydropower	0.011 (0.004)	0.011 (0.004)	0.009 (0.004)	0.009 (0.004)	0.006 (0.003)	0.006 (0.003)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.050	0.061	0.111	0.107	0.062	0.229
Bootstrap Wald	0.067	0.108	0.128	0.116	—	0.185
# Observations	2,808	2,808	2,808	2,808	2,808	2,457

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer 2015). All independent variables in the table enter the regression after being transformed into log changes. Rows under the “Controls for ...” heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Robust standard errors are shown in parentheses. Row “Wald” displays p -values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row “Bootstrap Wald” shows p -values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 5: TFP Growth vs. Renewable and Non-Renewable Natural Capital Growth

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Bauxite	0.002 (0.003)	0.002 (0.004)	0.001 (0.003)	0.001 (0.004)	0.000 (0.003)	-0.001 (0.003)
Coal	0.005 (0.004)	0.007 (0.004)	0.009 (0.004)	0.009 (0.004)	0.007 (0.003)	0.004 (0.003)
Oil	0.014 (0.007)	0.014 (0.007)	0.016 (0.007)	0.015 (0.007)	0.015 (0.008)	0.015 (0.008)
Copper	0.000 (0.003)	0.001 (0.003)	0.002 (0.003)	0.002 (0.003)	0.002 (0.003)	-0.002 (0.003)
Phosphate	0.014 (0.008)	0.014 (0.008)	0.015 (0.009)	0.015 (0.008)	0.015 (0.009)	0.017 (0.011)
Natural Gas	0.002 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.004)
Gold	0.002 (0.002)	0.002 (0.002)	0.003 (0.002)	0.003 (0.002)	0.002 (0.002)	0.002 (0.002)
Silver	0.001 (0.001)	0.001 (0.001)	0.002 (0.001)	0.002 (0.001)	0.002 (0.001)	0.001 (0.002)
Iron	0.002 (0.002)	0.003 (0.002)	0.001 (0.002)	0.002 (0.002)	0.003 (0.002)	0.004 (0.003)
Tin	0.002 (0.004)	0.002 (0.004)	0.004 (0.004)	0.004 (0.004)	0.004 (0.004)	-0.000 (0.005)
Lead	0.001 (0.002)	0.001 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)
Zinc	-0.000 (0.002)	-0.000 (0.002)	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.003)
Nickel	0.003 (0.003)	0.003 (0.003)	0.002 (0.003)	0.002 (0.003)	0.002 (0.003)	0.001 (0.003)
Urban Land	0.294 (0.080)	0.307 (0.093)	0.225 (0.061)	0.239 (0.068)	0.249 (0.074)	0.099 (0.078)
Agricultural Land	0.201 (0.102)	0.192 (0.076)	0.110 (0.085)	0.109 (0.084)	0.076 (0.067)	0.143 (0.098)
Timber Land	0.015 (0.025)	0.034 (0.025)	0.037 (0.023)	0.046 (0.023)	0.042 (0.021)	0.037 (0.018)
Forest Cover	-0.107 (0.105)	-0.171 (0.091)	-0.178 (0.095)	-0.206 (0.095)	-0.081 (0.090)	-0.084 (0.104)
Mangroves	0.094 (0.241)	0.099 (0.218)	0.149 (0.259)	0.172 (0.257)	0.333 (0.211)	0.157 (0.302)
Fishery Biomass	-0.101 (0.069)	-0.120 (0.058)	-0.020 (0.074)	-0.026 (0.074)	0.171 (0.106)	0.021 (0.103)
Hydropower	0.013 (0.005)	0.013 (0.005)	0.012 (0.005)	0.012 (0.005)	0.010 (0.005)	0.010 (0.004)
Wald	0.002	0.000	0.001	0.000	0.000	0.014
Bootstrap Wald	0.096	0.044	0.032	0.014	—	0.111
# Observations	5,255	5,255	5,255	5,255	5,255	4,901
# Countries	118	118	118	118	118	118

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer 2015). All independent variables in the table enter the regression after being transformed into log changes. Robust standard errors are shown in parentheses. Row “Wald” displays p -values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row “Bootstrap Wald” shows p -values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 6: TFP Growth vs. All NK Growth, 1996-2019 Sample

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Bauxite	0.003 (0.006)	0.003 (0.006)	0.001 (0.005)	0.001 (0.005)	-0.001 (0.004)	-0.001 (0.003)
Coal	0.004 (0.006)	0.006 (0.005)	0.002 (0.004)	0.003 (0.004)	0.003 (0.005)	0.001 (0.004)
Oil	0.003 (0.003)	0.003 (0.003)	0.003 (0.002)	0.002 (0.002)	0.002 (0.002)	0.003 (0.002)
Copper	-0.003 (0.003)	-0.003 (0.003)	-0.003 (0.003)	-0.003 (0.003)	-0.004 (0.003)	-0.007 (0.004)
Phosphate	0.018 (0.011)	0.017 (0.011)	0.021 (0.013)	0.020 (0.013)	0.020 (0.014)	0.020 (0.013)
Natural Gas	0.000 (0.001)	0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	0.000 (0.001)
Gold	0.003 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.001 (0.002)	0.001 (0.002)
Silver	0.002 (0.002)	0.002 (0.002)	0.003 (0.002)	0.003 (0.002)	0.003 (0.002)	0.002 (0.002)
Iron	0.004 (0.003)	0.004 (0.003)	0.004 (0.003)	0.004 (0.003)	0.005 (0.004)	0.007 (0.004)
Tin	-0.002 (0.002)	-0.001 (0.003)	-0.001 (0.002)	-0.001 (0.003)	-0.001 (0.003)	0.001 (0.003)
Lead	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.001 (0.002)
Zinc	-0.002 (0.003)	-0.002 (0.003)	-0.003 (0.002)	-0.003 (0.002)	-0.003 (0.002)	-0.004 (0.003)
Nickel	0.003 (0.005)	0.002 (0.005)	0.000 (0.004)	0.000 (0.004)	-0.000 (0.004)	-0.001 (0.003)
Urban Land	0.311 (0.092)	0.311 (0.097)	0.209 (0.065)	0.219 (0.073)	0.152 (0.060)	0.170 (0.064)
Agricultural Land	0.157 (0.090)	0.159 (0.076)	0.069 (0.073)	0.074 (0.071)	-0.009 (0.065)	0.147 (0.089)
Timber Land	0.028 (0.024)	0.044 (0.025)	0.032 (0.021)	0.038 (0.021)	0.044 (0.021)	0.036 (0.017)
Forest Cover	-0.067 (0.095)	-0.109 (0.091)	-0.014 (0.081)	0.004 (0.083)	0.083 (0.053)	0.075 (0.103)
Mangroves	0.147 (0.243)	0.091 (0.239)	0.357 (0.281)	0.327 (0.304)	0.315 (0.337)	0.166 (0.286)
Fishery Biomass	0.120 (0.094)	0.035 (0.074)	0.122 (0.111)	0.099 (0.104)	0.068 (0.129)	0.052 (0.107)
Hydropower	0.011 (0.005)	0.012 (0.005)	0.010 (0.004)	0.010 (0.004)	0.008 (0.004)	0.010 (0.004)
Wald	0.016	0.014	0.023	0.028	0.021	0.001
Bootstrap Wald	0.096	0.044	0.032	0.014	—	0.111
# Observations	2,832	2,832	2,832	2,832	2,832	2,802
# Countries	118	118	118	118	118	118

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer 2015). All independent variables in the table enter the regression after being transformed into log changes. Robust standard errors are shown in parentheses. Row “Wald” displays p -values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row “Bootstrap Wald” shows p -values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 7: EU27 + Norway Eurostat MFP Growth vs. Natural Capital Growth

Outcome: $\Delta \ln(MFP_{j,t})$	(1)	(2)	(3)	(4)	(5)
Timber Land	-0.477 (0.286)	-0.711 (0.252)	-0.818 (0.254)	-0.815 (0.246)	-1.147 (0.342)
Agricultural Land	0.027 (0.038)	0.030 (0.036)	0.035 (0.032)	0.041 (0.029)	0.035 (0.028)
Forest Cover	0.308 (0.373)	0.328 (0.329)	0.421 (0.315)	0.408 (0.292)	0.418 (0.311)
Biomass	-0.443 (0.107)	-0.522 (0.109)	0.268 (0.134)	0.198 (0.141)	0.243 (0.338)
Hydropower	-0.002 (0.004)	0.001 (0.004)	0.001 (0.004)	0.001 (0.004)	0.002 (0.004)
Controls for...					
Factor Inputs	No	Yes	No	Yes	Yes
TWFE	No	No	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes
Wald	0.001	0.000	0.006	0.005	0.023
Bootstrap Wald	0.015	0.003	0.115	0.096	—
# Observations	476	476	476	476	476
# Countries	28	28	28	28	28

Note: All independent variables in the table are measured in terms of log changes. The mangroves account is zero for all EU countries and is removed as a covariate for regressions in this table. Control rows denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. Robust standard errors are shown in parentheses. Row “Wald” displays p -values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row “Bootstrap Wald” shows p -values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 8: EU27 + Norway PWT TFP Growth vs. Natural Capital Growth

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)
Timber Land	-0.218 (0.189)	-0.306 (0.169)	-0.165 (0.155)	-0.148 (0.161)	-0.449 (0.258)
Agricultural Land	0.008 (0.035)	0.024 (0.033)	0.002 (0.029)	0.015 (0.025)	0.010 (0.025)
Forest Cover	0.191 (0.165)	0.192 (0.143)	0.172 (0.116)	0.174 (0.111)	0.103 (0.155)
Biomass	-0.275 (0.084)	-0.401 (0.099)	0.023 (0.122)	0.052 (0.137)	0.019 (0.298)
Hydropower	-0.004 (0.004)	-0.000 (0.004)	0.000 (0.004)	0.001 (0.003)	0.001 (0.003)
Controls for . . .					
Factor Inputs	No	Yes	No	Yes	Yes
TWFE	No	No	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes
Wald	0.020	0.002	0.807	0.702	0.405
Bootstrap Wald	0.095	0.004	0.828	0.752	—
# Observations	476	476	476	476	476
# Countries	28	28	28	28	28

Note: All independent variables in the table are measured in terms of log changes. The mangroves account is zero for all EU countries and is removed as a covariate for regressions in this table. Control rows denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. Robust standard errors are shown in parentheses. Row “Wald” displays p -values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row “Bootstrap Wald” shows p -values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 9: Agricultural TFP Growth vs. Natural Capital Growth

Outcome: $\Delta \ln(Ag. TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)
Timber Land	0.164 (0.049)	-0.131 (0.186)	-0.067 (0.093)	-0.284 (0.266)	-0.374 (0.361)
Agricultural Land	0.027 (0.019)	0.037 (0.049)	0.059 (0.037)	0.039 (0.054)	0.057 (0.059)
Forest Cover	-0.067 (0.061)	0.157 (0.200)	0.078 (0.124)	0.221 (0.247)	0.276 (0.285)
Mangroves	0.122 (0.128)	-0.108 (0.327)	0.247 (0.317)	0.426 (0.505)	0.736 (0.595)
Biomass	0.112 (0.056)	0.072 (0.140)	-0.024 (0.153)	-0.055 (0.171)	0.005 (0.225)
Hydropower	0.011 (0.005)	-0.012 (0.008)	-0.007 (0.008)	-0.015 (0.009)	-0.016 (0.009)
Controls for...					
Factor Inputs	No	Yes	No	Yes	Yes
TWFE	No	No	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes
Wald	0.001	0.769	0.572	0.495	0.275
Bootstrap Wald	0.001	0.829	0.606	0.582	–
# Observations	2,808	2,377	3,399	2,377	2,377
# Countries	117	114	163	114	114

Note: All independent variables in the table are measured in terms of log changes. Control rows denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. Row “Wald” displays p -values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row “Bootstrap Wald” shows p -values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 10: National Accounts-Based Real GDP Growth vs. Natural Capital Growth

Outcome: $\Delta \ln(Y_{j,t})$	(1)	(2)	(3)	(4)	(5)
Timber Land	0.000 (0.092)	0.121 (0.105)	-0.079 (0.114)	-0.097 (0.118)	-0.092 (0.198)
Agricultural Land	0.067 (0.050)	0.091 (0.050)	0.096 (0.052)	0.101 (0.055)	0.098 (0.071)
Forest Cover	0.278 (0.133)	0.038 (0.132)	0.111 (0.164)	0.135 (0.174)	-0.060 (0.166)
Mangroves	-0.015 (0.265)	0.148 (0.255)	0.650 (0.337)	0.625 (0.452)	1.271 (0.647)
Fishery Biomass	-0.051 (0.125)	0.044 (0.111)	0.108 (0.133)	-0.008 (0.172)	-0.280 (0.303)
Hydropower	0.010 (0.006)	0.010 (0.006)	0.006 (0.007)	0.002 (0.008)	-0.001 (0.007)
Controls for...					
Other Factor Inputs	No	Yes	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	Yes	No
Wald	0.033	0.000	0.024	0.181	0.017
Bootstrap Wald	0.063	0.000	0.014	—	0.003
# Observations	3,258	2,808	2,808	2,808	2,457
# Countries	136	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $RGDP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer 2015). All independent variables in the table are measured in terms of log changes. Other factor input controls include the human capital and labor share variables in the PWT. Other controls allow for country and year fixed effects (TWFE) and idiosyncratic time trends at the country level. Row “Wald” displays p -values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row “Bootstrap Wald” shows p -values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (4) due to the idiosyncratic trend terms.

Table 11: PPP-Based Output Growth vs. Natural Capital Growth

Outcome: $\Delta \ln(Y_{j,t})$	(1)	(2)	(3)	(4)	(5)
Timber Land	0.054 (0.054)	0.189 (0.048)	0.095 (0.049)	0.015 (0.054)	-0.092 (0.198)
Agricultural Land	0.021 (0.017)	0.045 (0.019)	0.035 (0.018)	0.039 (0.018)	0.098 (0.071)
Forest Cover	0.011 (0.065)	-0.176 (0.066)	-0.025 (0.063)	0.064 (0.058)	-0.060 (0.166)
Mangroves	0.125 (0.108)	0.080 (0.116)	0.281 (0.159)	0.262 (0.200)	1.271 (0.647)
Fishery Biomass	-0.008 (0.053)	0.007 (0.053)	0.056 (0.060)	0.027 (0.082)	-0.280 (0.303)
Hydropower	0.011 (0.004)	0.011 (0.004)	0.008 (0.005)	0.006 (0.005)	-0.001 (0.007)
Controls for...					
Other Factor Inputs	No	Yes	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	Yes	No
Wald	0.091	0.178	0.159	0.342	0.307
Bootstrap Wald	0.164	0.210	0.195	—	0.371
# Observations	3,264	2,808	2,808	2,808	2,457
# Countries	136	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $CGDP^O$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer 2015). All independent variables in the table are measured in terms of log changes. Other factor input controls include the human capital and labor share variables in the PWT. Other controls allow for country and year fixed effects (TWFE) and idiosyncratic time trends at the country level. Row “Wald” displays p -values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row “Bootstrap Wald” shows p -values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (4) due to the idiosyncratic trend terms.

Table 12: Non-renewable resources used to construct empirical N_1

Resource	Unit
Coal	Metric tons
Gas	Terajoules
Oil	Barrels
Bauxite	Metric tons
Copper	Metric tons
Gold	Metric tons
Iron	Metric tons
Lead	Metric tons
Nickel	Metric tons
Phosphate	Metric tons
Silver	Metric tons
Tin	Metric tons
Zinc	Metric tons

Table 13: Renewable resources used to construct empirical N_2

Resource	Unit
Productive timber forest area	Square Kilometers
Agricultural land area	Square Kilometers
Non-timber forest cover	Square Kilometers
Mangrove ecosystem Area	Hectares
Fishery biomass	Metric tons
Hydroelectric Power Generation	Gigawatt-hours

Table 14: Output Growth vs. Natural Capital Growth

$\Delta \ln(y_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
$g_{j,t}^R$	0.078*** (0.021)	0.046** (0.020)	0.052*** (0.020)	0.042** (0.019)	0.046** (0.020)	0.035** (0.014)
$g_{j,t}^{NR}$	0.025*** (0.004)	0.013*** (0.003)	0.017*** (0.004)	0.011*** (0.003)	0.011*** (0.003)	0.013*** (0.004)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
# Observations	4974	3953	4974	3953	3953	3697
# Countries	134	104	134	104	104	104

Note: Robust standard errors are shown in parentheses. Rows under the “Controls for ...” heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Stars ** indicate $p < 0.05$, *** indicate $p < 0.01$.

Table 15: Simulation Moments and Data Sources

Description	Population Moment	Sample Moment	Source
Growth in Non-Renewable Capital Services	$\mathbb{E}[g_{N_1}]$	$\bar{g}^R$	CWON
Growth in Renewable Capital Services	$\mathbb{E}[g_{N_2}]$	$\bar{g}^{NR}$	CWON
Growth in Produced Capital Services	$\mathbb{E}[g_K]$	$\bar{g}_K$	PWT
Growth in Labor Inputs	$\mathbb{E}[g_L]$	$\bar{g}_L$	PWT
Growth in TFP	$\mathbb{E}[g_{TFP}]$	$\bar{g}_{TFP}$	PWT
Covariances of Growth Terms	Σ	$\hat{\Sigma}$	CWON & PWT

Note: Subscripts indexing each object by country are dropped in the first two columns; all population moments are assigned the sample analogs on a country-by-country basis. Bars denote sample averages while hats denote sample analogs for the covariance matrix Σ . CWON refers to the World Bank *Changing Wealth of Nations* database ([World Bank 2025](#)). PWT refers to the Penn World Tables version 10.01 (Feenstra, Inklaar, and Timmer [2015](#)).

Table 16: Country-Level Sample Moments and Simulation Results

ISO3 Code	Country	$\bar{g}_{N_1}$	$\bar{g}_{N_2}$	$\hat{\rho}(g_{N_1}, g_K)$	$\hat{\rho}(g_{N_1}, g_{N_2})$	Bias Reduction	RMSE Reduction	$\bar{g}_{TFP}$
AGO	Angola	0.04	0.01	0.13	0.07	9.25e-04	1.02e-04	0.02
ARG	Argentina	-0.01	-0.00	0.13	-0.15	8.44e-04	5.29e-05	-0.00
ARM	Armenia	0.16	0.01	0.16	0.19	9.03e-03	2.90e-03	0.06
AUS	Australia	0.05	-0.01	-0.13	0.08	2.75e-03	1.55e-03	0.00
AUT	Austria	-0.02	0.00	-0.08	0.09	-1.78e-04	3.05e-06	0.00
BDI	Burundi	0.08	-0.00	0.04	0.03	4.94e-03	2.09e-03	-0.01
BEL	Belgium	0.22	-0.00	0.12	-0.56	3.99e-03	1.05e-03	0.00
BEN	Benin	-0.04	0.01	-0.16	-0.04	2.76e-03	6.61e-04	0.02
BFA	Burkina Faso	0.21	0.00	0.06	-0.10	2.09e-03	4.25e-04	0.01
BGR	Bulgaria	-0.01	-0.00	-0.27	0.14	1.26e-03	9.79e-05	-0.01
BHR	Bahrain	0.04	-0.00	0.13	-0.00	2.07e-03	7.52e-04	-0.01
BOL	Bolivia	0.04	-0.00	-0.19	-0.12	1.01e-03	1.86e-04	0.00
BRA	Brazil	0.06	0.01	0.15	0.08	1.29e-03	2.94e-04	-0.01
BWA	Botswana	0.01	-0.00	-0.08	-0.16	4.01e-03	9.23e-04	-0.01
CAF	Central African Republic	0.08	0.00	0.31	-0.15	3.95e-03	4.07e-04	-0.00
CAN	Canada	0.02	0.00	0.13	-0.04	4.30e-04	6.54e-05	0.00
CHE	Switzerland	0.15	0.00	-0.03	-0.51	2.63e-03	1.55e-03	0.01
CHL	Chile	0.03	0.00	0.30	-0.42	-1.42e-03	-3.35e-04	-0.00
CHN	China	0.04	0.00	0.54	-0.01	7.86e-03	4.37e-03	0.01
CIV	Côte d'Ivoire	0.09	0.00	-0.12	-0.28	2.04e-03	3.20e-04	0.01
CMR	Cameroon	0.02	0.00	-0.24	0.45	1.69e-02	1.33e-02	0.01
COL	Colombia	0.02	0.00	0.40	0.04	9.08e-04	1.77e-04	-0.00
CRI	Costa Rica	0.18	-0.00	-0.15	-0.03	2.18e-02	1.62e-02	0.01
CZE	Czech Republic	-0.01	-0.00	0.22	-0.43	1.26e-03	2.23e-04	0.01
DEU	Germany	-0.03	-0.00	0.37	-0.14	3.82e-03	2.10e-03	0.00
DNK	Denmark	-0.03	-0.00	0.31	-0.55	1.09e-02	8.08e-03	0.00
DOM	Dominican Republic	-0.01	-0.00	0.04	-0.21	3.61e-03	1.05e-03	0.02
ECU	Ecuador	0.01	0.01	0.01	-0.20	-3.96e-04	-4.10e-05	0.00
EGY	Egypt	-0.00	0.00	0.09	0.38	1.36e-04	2.46e-05	-0.01
ESP	Spain	-0.02	-0.00	-0.49	0.05	8.95e-03	7.23e-03	-0.00
EST	Estonia	0.06	0.00	0.24	0.68	2.90e-03	6.21e-04	0.02
FIN	Finland	0.07	-0.00	0.19	0.28	1.38e-03	3.56e-04	0.01
FRA	France	-0.07	-0.00	0.04	0.19	4.21e-03	2.60e-03	0.00
GAB	Gabon	-0.03	-0.00	0.10	-0.36	4.76e-04	1.76e-05	-0.01
GBR	United Kingdom	-0.05	0.00	0.46	-0.20	7.51e-03	5.76e-03	0.00
GRC	Greece	-0.02	-0.01	0.20	-0.06	2.06e-03	6.20e-04	-0.00
GTM	Guatemala	0.02	-0.00	-0.03	0.08	9.05e-03	5.60e-03	0.00
HND	Honduras	0.02	-0.00	-0.13	0.02	2.07e-03	3.71e-04	-0.00
HRV	Croatia	-0.01	-0.01	0.07	-0.04	2.85e-03	1.17e-03	0.01
HUN	Hungary	-0.04	-0.00	-0.06	0.11	1.69e-03	4.54e-04	0.01
IDN	Indonesia	0.01	0.01	0.16	-0.06	-9.80e-05	-7.29e-06	0.00
IND	India	0.03	-0.00	-0.03	-0.01	9.94e-04	1.35e-04	0.02
IRL	Ireland	-0.03	0.00	-0.03	0.10	6.23e-04	2.49e-05	0.01
IRN	Iran	0.00	-0.01	0.20	-0.08	1.95e-03	2.33e-04	-0.01
IRQ	Iraq	0.10	0.00	-0.34	-0.23	3.63e-03	2.31e-04	0.03

ISO3 Code	Country	$\bar{g}_{N_1}$	$\bar{g}_{N_2}$	$\hat{\rho}(g_{N_1}, g_K)$	$\hat{\rho}(g_{N_1}, g_{N_2})$	Bias Reduction	RMSE Reduction	$\bar{g}_{TFP}$
ISR	Israel	0.10	0.00	-0.59	-0.31	2.60e-02	2.23e-02	0.00
ITA	Italy	-0.04	-0.00	-0.07	-0.05	1.51e-03	5.50e-04	-0.01
JAM	Jamaica	-0.00	-0.00	0.35	-0.18	2.08e-03	5.72e-04	-0.01
JOR	Jordan	0.03	-0.00	-0.04	-0.10	1.46e-03	2.98e-04	-0.00
JPN	Japan	-0.02	-0.00	-0.10	-0.24	4.12e-04	5.22e-05	0.00
KAZ	Kazakhstan	0.02	0.00	-0.05	-0.22	2.52e-03	4.24e-04	0.03
KEN	Kenya	0.15	-0.00	-0.10	-0.00	5.08e-02	4.41e-02	-0.00
KGZ	Kyrgyzstan	0.08	-0.00	-0.38	0.22	1.90e-02	1.31e-02	0.02
KOR	South Korea	-0.08	-0.01	0.06	0.19	7.48e-03	4.44e-03	0.01
KWT	Kuwait	0.03	0.00	0.07	0.21	1.32e-03	8.84e-05	-0.03
LAO	Laos	0.19	0.01	-0.04	0.33	3.22e-02	2.72e-02	0.00
LKA	Sri Lanka	0.01	0.00	0.14	0.20	3.91e-03	1.91e-03	0.00
LTU	Lithuania	0.00	-0.00	-0.19	-0.46	3.32e-03	8.30e-04	0.02
MAR	Morocco	0.01	-0.00	-0.29	-0.14	9.81e-03	5.69e-03	0.01
MEX	Mexico	-0.01	-0.00	0.16	0.36	2.05e-03	6.52e-04	-0.00
MNG	Mongolia	0.09	-0.00	-0.06	0.15	3.61e-03	1.03e-03	0.02
MOZ	Mozambique	0.14	0.02	0.06	-0.09	1.47e-02	1.00e-02	0.01
MRT	Mauritania	-0.00	-0.00	0.14	0.45	2.74e-03	4.20e-04	-0.00
MYS	Malaysia	0.01	0.00	0.50	0.08	7.77e-04	7.06e-05	0.01
NAM	Namibia	0.02	-0.00	-0.05	-0.03	1.11e-03	1.38e-04	-0.00
NER	Niger	0.01	0.01	0.06	0.17	2.49e-02	1.84e-02	0.01
NGA	Nigeria	0.01	0.00	0.40	0.01	1.84e-03	2.48e-04	0.02
NIC	Nicaragua	0.11	-0.00	-0.18	-0.09	1.06e-02	7.25e-03	-0.00
NLD	Netherlands	-0.03	-0.00	0.02	0.05	3.29e-03	1.80e-03	0.00
NOR	Norway	0.00	-0.00	-0.02	-0.06	5.64e-03	3.59e-03	-0.00
NZL	New Zealand	0.01	-0.01	0.08	0.07	8.75e-04	2.81e-04	0.00
PAN	Panama	0.12	0.01	-0.00	-0.33	1.73e-03	1.94e-04	-0.00
PER	Peru	0.05	0.01	-0.39	-0.12	7.38e-03	4.95e-03	-0.00
PHL	Philippines	0.07	0.00	-0.26	-0.09	9.68e-03	6.82e-03	0.01
POL	Poland	0.00	-0.00	0.11	0.18	9.37e-04	2.55e-04	0.02
PRT	Portugal	-0.00	-0.00	-0.28	-0.04	2.10e-03	8.97e-04	-0.00
QAT	Qatar	0.06	0.00	0.32	0.02	-3.67e-04	-9.50e-06	-0.03
ROU	Romania	-0.06	-0.00	0.13	-0.22	3.80e-03	9.17e-04	0.02
RUS	Russia	-0.03	-0.00	-0.02	-0.08	1.16e-03	1.02e-04	0.02
RWA	Rwanda	0.18	0.01	0.16	-0.51	1.03e-02	5.34e-03	0.03
SAU	Saudi Arabia	0.07	0.00	0.07	-0.29	2.40e-03	7.21e-04	-0.02
SDN	Sudan	0.14	-0.01	0.25	0.52	4.24e-03	1.45e-03	0.01
SEN	Senegal	0.07	-0.00	0.32	0.10	5.50e-03	2.51e-03	0.00
SLE	Sierra Leone	0.14	0.00	0.17	0.03	1.02e-02	2.25e-03	-0.00
SRB	Serbia	-0.03	-0.00	0.25	-0.28	3.59e-03	6.91e-04	0.03
SVK	Slovakia	-0.04	-0.00	0.03	-0.06	2.25e-03	6.75e-04	0.02
SVN	Slovenia	-0.03	0.01	-0.05	-0.35	9.21e-04	1.03e-04	0.01
SWE	Sweden	0.03	-0.00	0.11	0.20	-3.12e-05	1.74e-05	0.01
SWZ	Eswatini	0.03	0.00	0.15	-0.11	5.01e-03	1.86e-03	0.01
TGO	Togo	0.09	0.00	-0.25	0.18	1.78e-02	1.09e-02	0.02
THA	Thailand	0.05	0.00	0.32	-0.24	7.36e-04	4.03e-05	0.02
TTO	Trinidad and Tobago	0.01	-0.01	-0.05	0.17	4.76e-04	1.21e-05	0.00

ISO3 Code	Country	$\bar{g}_{N_1}$	$\bar{g}_{N_2}$	$\hat{\rho}(g_{N_1}, g_K)$	$\hat{\rho}(g_{N_1}, g_{N_2})$	Bias Reduction	RMSE Reduction	$\bar{g}_{TFP}$
TUN	Tunisia	-0.01	0.00	0.34	-0.06	4.78e-03	2.94e-03	0.01
TUR	Turkey	0.04	0.00	0.06	0.22	1.31e-03	1.82e-04	-0.00
TZA	Tanzania	0.12	0.00	-0.43	0.07	2.40e-02	2.02e-02	0.01
URY	Uruguay	-0.07	0.02	0.08	-0.14	4.51e-03	2.60e-03	0.01
USA	United States	0.02	-0.00	-0.39	-0.23	3.69e-03	2.58e-03	0.01
ZAF	South Africa	0.00	-0.00	0.24	0.31	1.33e-03	2.50e-04	-0.00
ZMB	Zambia	0.04	0.00	0.08	0.23	1.64e-03	2.66e-04	0.02
ZWE	Zimbabwe	0.00	0.01	0.32	0.40	-6.74e-04	8.91e-07	0.01

Table 17: RMSE Change Decomposition

Dep. Var: ΔRMSE	(1)
$\rho(g_{N_1}, g_K)$	0.006 (0.003)
$\rho(g_{N_1}, g_{N_2})$	-0.003 (0.002)
$\rho(g_{N_2}, g_L)$	0.002 (0.003)
$\rho(g_{N_1}, g_L)$	0.008 (0.003)
$\rho(g_L, g_K)$	0.004 (0.002)
$\rho(g_{N_2}, g_K)$	-0.002 (0.003)
$\bar{g}_L$	0.016 (0.057)
$\bar{g}_{N_1}$	-0.039 (0.010)
$\bar{g}_K$	0.001 (0.034)
$\bar{g}_{N_2}$	-0.164 (0.090)
$\bar{g}_A$	0.040 (0.059)
σ_{TFP}	0.058 (0.025)
R^2	0.38
N	100

Heteroskedasticity-robust standard errors shown in parentheses.

8 Figures

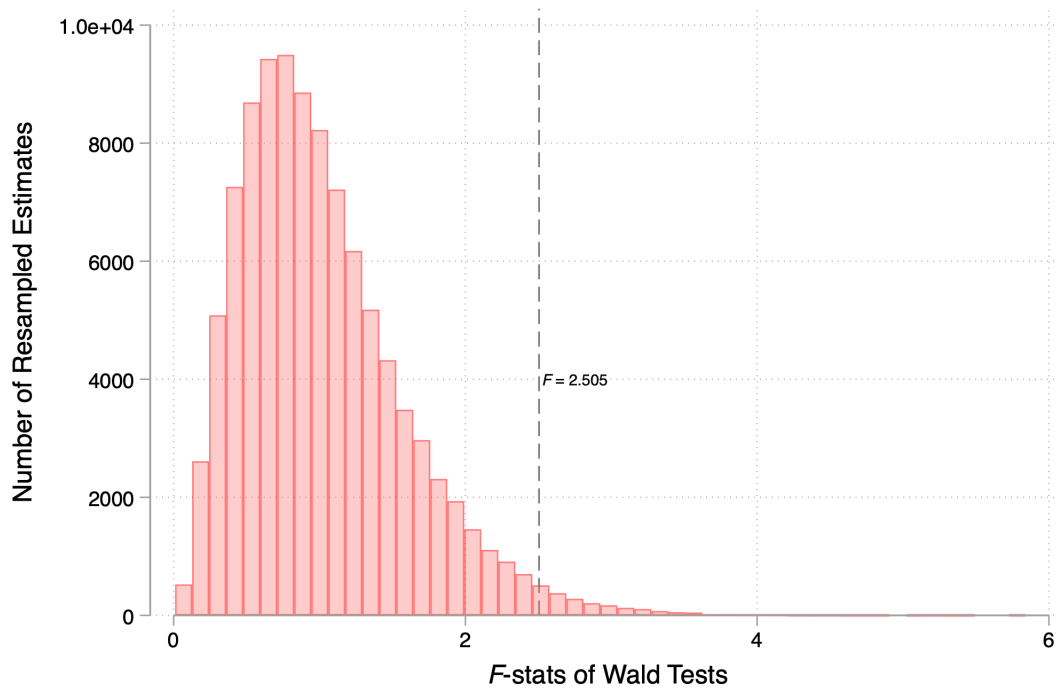


Figure 1: Histogram of F -statistics from Wald tests of joint nullity performed on 100,000 permuted samples of the data where the null is imposed true in expectation.

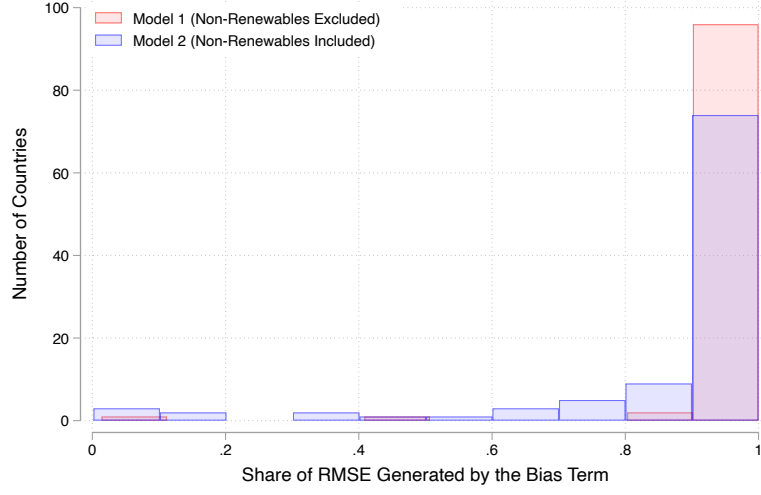


Figure 2: Histograms showing the share of the RMSE terms under each model attributed to the (squared) bias terms $\mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}]]^2$ and $\mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}]]^2$.

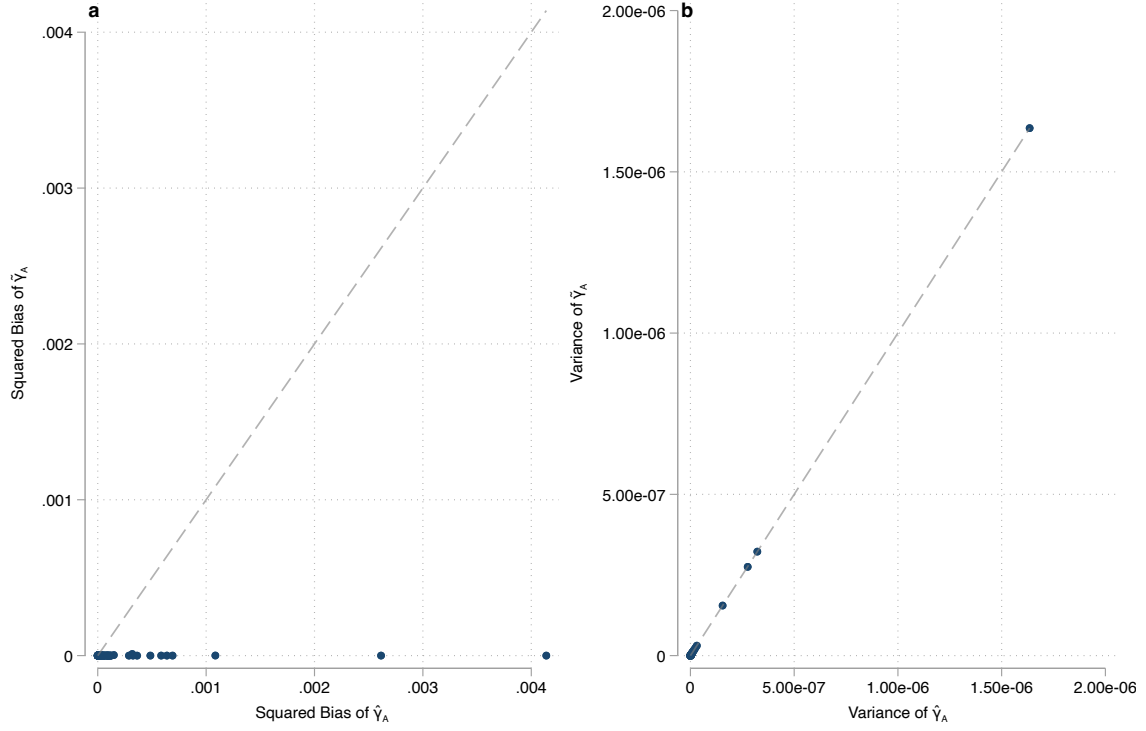


Figure 3: Panel 3a shows country-level biases in $\hat{\gamma}_A$ and $\tilde{\gamma}_A$. Panel 3b shows their variances. Dashed lines are drawn along the 45-degree line where the squared biases or variances are equal.

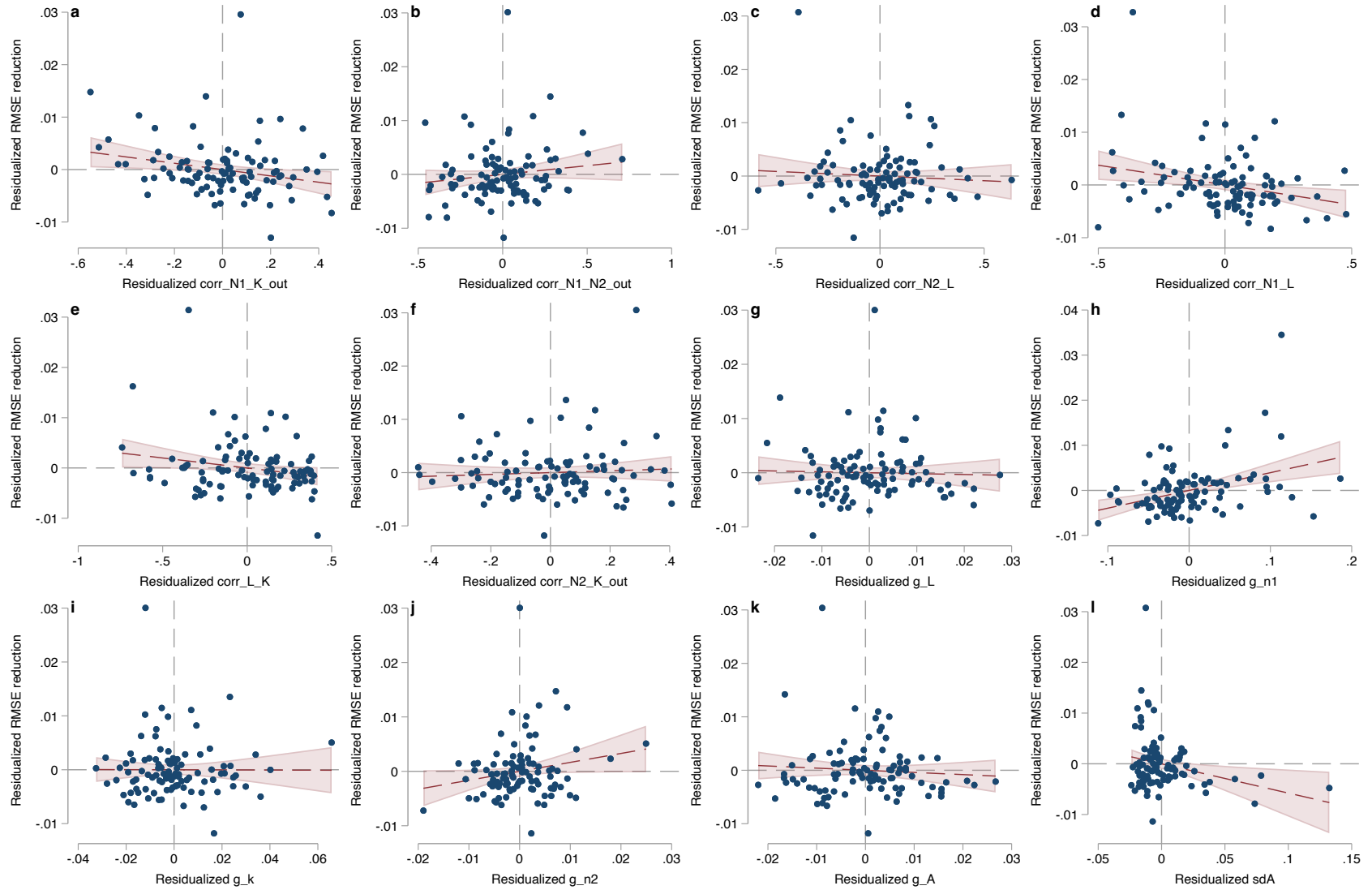


Figure 4: Frisch-Waugh-Lovell plot of residualized ΔRMSE against residualized versions of each of the 12 sample moments that determine its value at the country level. Dashed black lines show the linear regression coefficients in Table 17 while the red shaded regions denote confidence intervals of the fitted values.

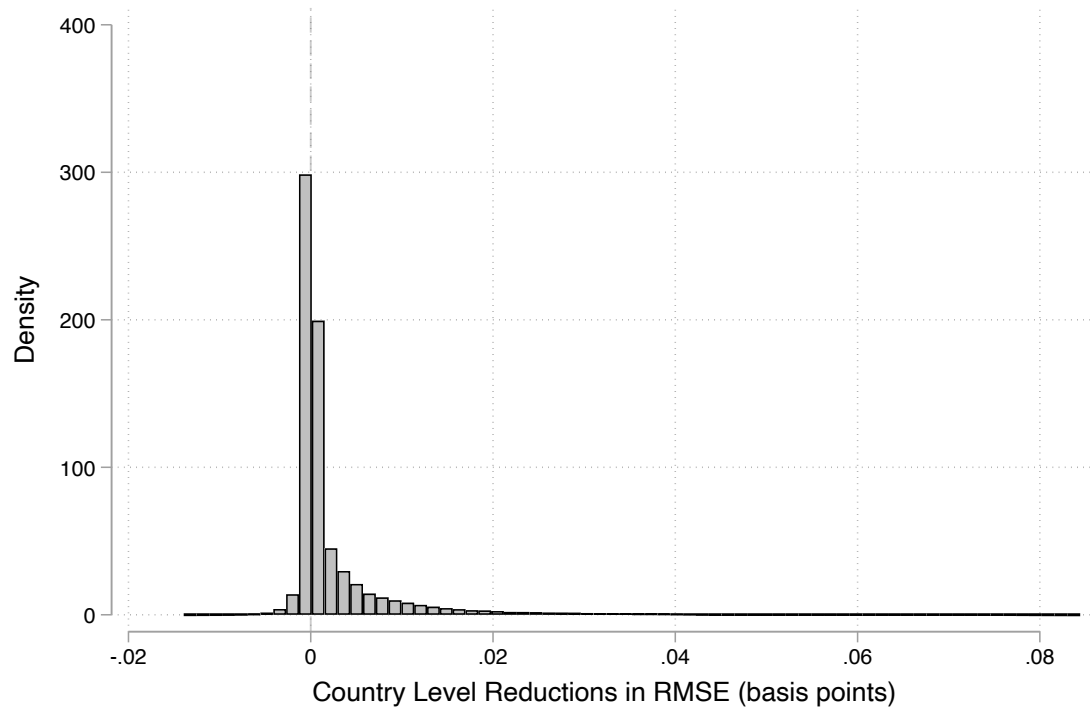


Figure 5: Histogram of ΔRMSE_j across each of $N = 10,000,000$ Monte Carlo draws of alternative output elasticity pairs γ_1, γ_2 at the country level.

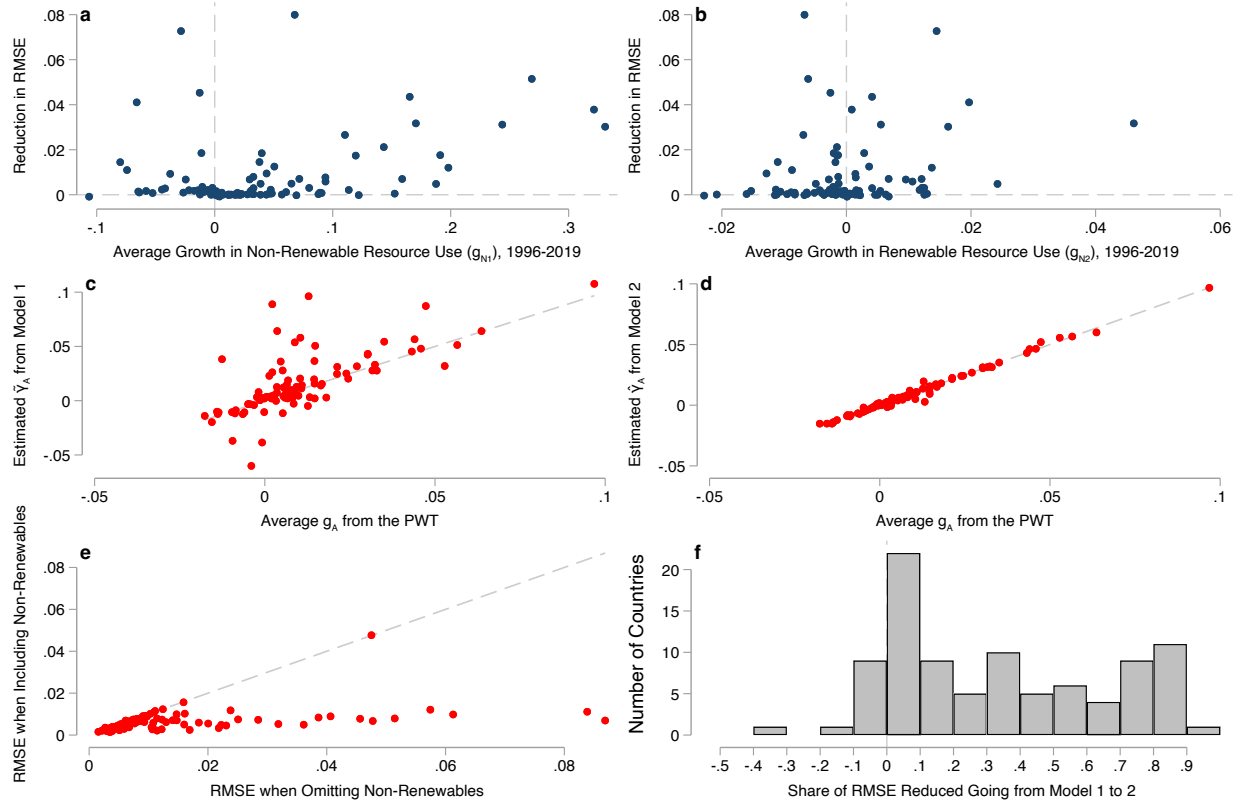


Figure 6: Panels 6a and 6b plot improvements in estimates of TFP growth against average growth in non-renewable and renewable resource use at the country level between 1996 and 2008 for 98 countries out of our original panel of 100 countries. Panels 6c and 6d show point estimates of TFP growth based on historical moments from the two estimators against the true data-generating process. Panels 6e and 6f show the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. Dashed lines in panels 6a, 6b, and 6f are plotted at $x = 0$ and $y = 0$. Dashed lines in panels 6c-6e show the 45-degree line.

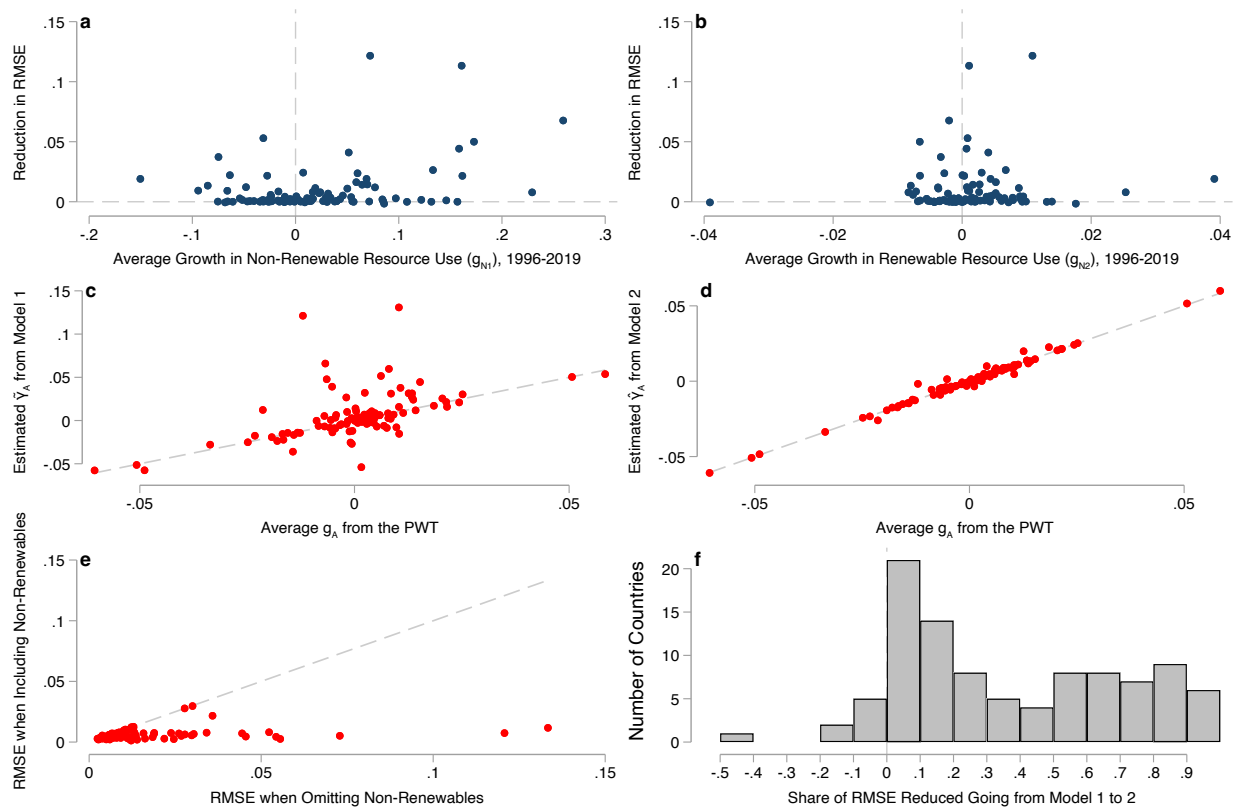


Figure 7: Panels 7a and 7b plot improvements in estimates of TFP growth against average growth in non-renewable and renewable resource use at the country level between 1996 and 2019 for 98 countries out of our original panel of 100 countries. Panels 7c and 7d show point estimates of TFP growth based on historical moments from the two estimators against the true data-generating process. Panels 7e and 7f show the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. Dashed lines in panels 7a, 7b, and 7f are plotted at $x = 0$ and $y = 0$. Dashed lines in panels 7c-7e show the 45-degree line.

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